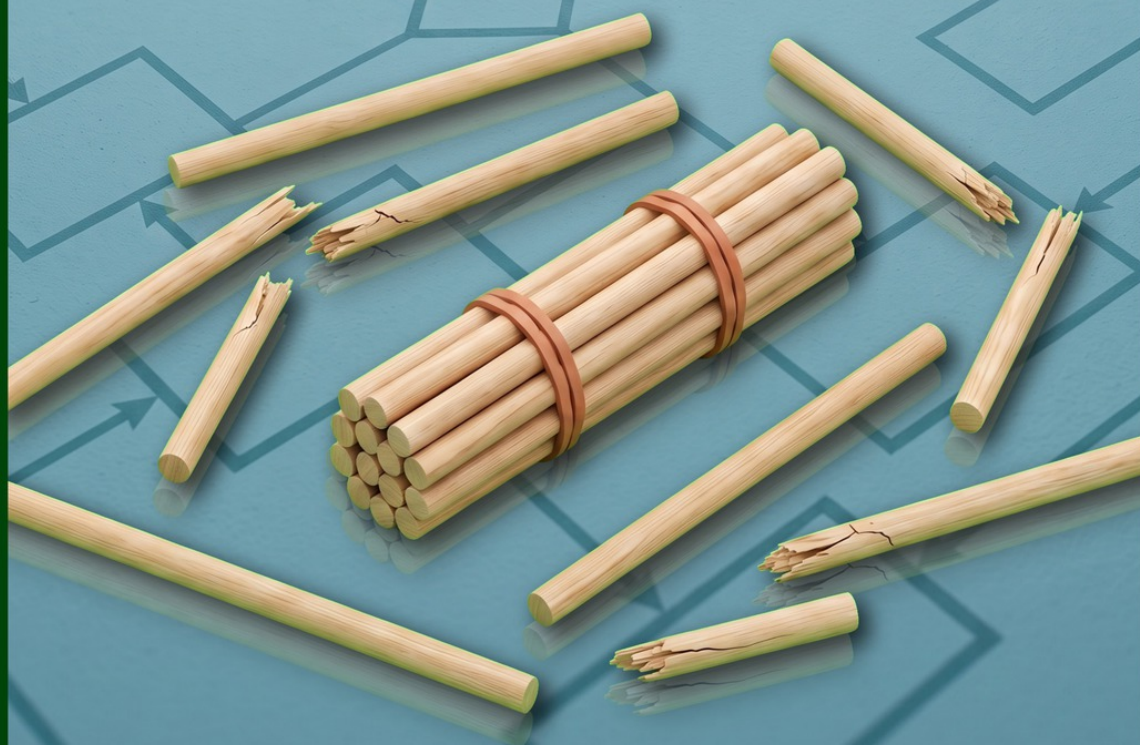


Antonio Ferrão Neto

THE METAPHOR OF THE BUBBLELAND

A PROTOSCIENTIFIC APPROACH

Appendices: Foundations and Hypotheses



Volume II

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APPENDIX **A**

The scientific method and protoscience

Science is not just logic; it is also history and tradition.

— THOMAS KUHN

*Scientific methodology is rich in diversity and cannot be
reduced to a single rule.*

— PAUL FEYERABEND

WE CAN understand science as the collective effort to build rational, testable, and revisable explanations about reality, with the aim of understanding, predicting, and transforming the world.

1. Why two appendices on method and hypotheses?

The presentation of the scientific method and of the types of hypotheses, with a level of detail uncommon in technical works, serves three articulated objectives:

1. **To make crystal clear the positioning of the work as protoscientific** relative to the *status quo* of established science, preventing legitimate weak hypotheses from being confused with consolidated scientific claims — or, at the opposite extreme, with disguised pseudoscience.
2. **To establish public and auditable criteria of admissibility**, so that each protoscientific hypothesis presented in the remaining chapters can be critically evaluated according to explicit rules, and not by the mere authority of the author. These criteria are developed in Appendix B, Section B.2 (**Cr1–Cr5**, individual) and Section B.3 (**Cf1–Cf8**, collective).
3. **To highlight the importance of publishing hypotheses not yet confirmed and falsified hypotheses**, both today underrepresented in the formal scientific literature (see Section A.18.1 of this appendix and the following ones).

Suggested reading itinerary.

- Appendix A (**Scientific Method**): situates the position of the work within the classical flowchart of the method — specifically, at the stage of *stating hypotheses*.
- Appendix B (**Types of Hypotheses**): presents the taxonomy, the criteria of individual admissibility (**Cr**) and those of collective strengthening (**Cf**).
- Appendix AH (Vol. III) (**Mapping of Hypotheses**): applies the criteria to the 46 hypotheses of the book, grouping them into six mutually reinforcing clusters.

The reader who wishes to evaluate the work on its own terms may regard these three appendices as the *methodological apparatus* that makes the rest of the book auditable — not merely readable.

Despite its apparent simplicity, this definition is only a starting point. Philosophers such as Alan Chalmers remind us that the term “scientific” is used to confer on a statement or theory a special merit of reliability, yet there is no consensus on what, in fact, the method or the criterion is that justifies this reliability (CHALMERS, 1999). Thus, understanding what science is and how it distinguishes itself from other forms of knowledge requires examining diverse historical and philosophical conceptions, such as *inductivism*, *falsificationism*, and the views of Kuhn, Lakatos, and Feyerabend.

In the following topics, we will discuss these conceptions, highlighting their contributions and limitations, so as to build a critical view of the so-called “scientific method”. As Locke argued, dogmatic errors — cognitive and institutional — prevent an inquiry from remaining open to criticism and to experience (Section A.3).

A.1 Tradition, terminology, and structure: when the name is not the thing

If science is also history and tradition, as Kuhn observed, then its language cannot be regarded as neutral or timeless. Terms, labels, and eponyms are products of cultural trajectories, educational institutions, and evaluation systems. This fact, apparently peripheral, has relevant epistemological implications.

Structure versus designation

A scientific structure — a coherent set of relations describing some aspect of reality — is not identical with the name that designates it. The structure has internal coherence, explanatory power, and observable consequences; the name is contingent. Formulas, laws, and principles may receive distinct labels in different national, pedagogical, or linguistic traditions, without this altering their content in any way. The same mathematical or physical object may be taught under various names: the structure is invariant, but the designation is a historical product.

Institutional repetition, however, transforms a terminological choice into a consolidated cultural standard. In psychology, this mechanism is called the *illusory truth effect* (HASHER; GOLDSTEIN; TOPPINO, 1977):¹ the repetition of a statement increases the perception that it is true, regardless of its truth value.

Revisability of terminology

Scientific terminology is itself revisable — just as are the theories it designates. When new historical evidence or new conventions of the scientific community require it, established names can be officially changed, without this representing any change in the underlying structure. A notable example is Hubble's Law, officially renamed the *Hubble–Lemaître Law* by the International Astronomical Union in 2018 (International Astronomical Union, 2018), in recognition of Georges Lemaître's prior contribution to the formulation of the relation between recession velocity and the distance of galaxies. The physical relation remained identical; only the label was

¹ There is a phrase wrongly attributed to Joseph Goebbels, propaganda minister of the Third Reich: "Repeat a lie a thousand times and it becomes the truth". As Bytwerk (2006) demonstrated, there is no record of this phrase in the minister's writings or speeches; the quotation seems to have arisen on the Anglophone internet without a primary source and to have spread through uncritical repetition, even reaching peer-reviewed journals (O'BRIEN, 2010). The irony is that the apocryphal quotation itself proves, through its diffusion, the psychological mechanism it purports to describe.

corrected. The community itself recognizes, in doing so, that the name is a cultural artifact superimposed on the structure, and not part of it.

Epistemological consequence

To confuse the name with the structure it designates is to commit a *category error*: to attribute ontological status to a cultural marker. Epistemological maturity requires distinguishing three planes:

1. **linguistic conventions** — the labels, always contingent;
2. **pedagogical traditions** — national or scholastic emphases in the attribution of authorship;
3. **invariant structures** — that which the theory effectively describes about the world.

The label is historical; the structure is, by hypothesis, independent of the history that named it. This distinction, far from being merely terminological, is also communicational: when scientific language is carried into didactic, popular, or fictional contexts, the distance between name and structure tends to grow — and with it, the risk that the reader confuses one with the other.

A.2 Hypothesis on intelligence, cognitive effort, and preference for *hard Sci-fi*: formulation and test plan

This section presents an example of applying the scientific method starting from an intuitive hypothesis: **the possible existence of a correlation between cognitive performance (often summarized under the label of “intelligence”) and preference for narratives classified as *Hard Sci-Fi***. This is a speculative proposition that requires rigorous operational definition, systematic data collection, and appropriate statistical analysis before any conclusive claim.

A.2.1 Motivation and practical relevance

Hard Sci-Fi productions tend to involve high costs and greater commercial risk, owing to factors such as:

- **Computer-generated imagery (CGI) and visual-effects budget**: representing technically plausible and visually sophisticated settings entails high investment.

- **Specialized scriptwriting:** maintaining internal scientific consistency demands research, consulting, and technical elaboration.
- **Demand for sustained attention:** narratives with high conceptual density may compete unfavorably with fragmented, multitasking consumption habits.

Several productions with more technical or scientifically structured pretensions were canceled after commercial performance below expectations, frequently associated with low return on investment (ROI). Among the examples frequently cited in public debates about audience and economic viability are: *Debris*, *Counterpart*, *The Peripheral*, *Constellation*, *Scavengers Reign*, *Raised by Wolves*, *Westworld*, *Almost Human*, and *Stargate Universe*.

By contrast, productions that favor mythology, character charisma, or narrative solutions less committed to physical-scientific consistency frequently achieve a larger mass audience. Examples discussed in this context include *Star Wars*, *Doctor Who*, *Manifest*, *Star Trek: Discovery*, *La Brea*, *Outlander*, and *Lost*.

The relevant scientific question does not involve aesthetic judgment, but the investigation of possible measurable cognitive determinants associated with narrative preferences.

A.2.2 Formulation of the hypothesis

Main hypothesis (H1).

There is a positive correlation between performance on standardized measures of cognitive ability and preference for *Hard Sci-Fi*.

Null hypothesis (H0).

There is no statistically significant correlation between cognitive performance and preference for *Hard Sci-Fi*, after controlling for relevant confounding variables.

Rival hypotheses.

- H2: The observed association is mediated by traits such as intellectual curiosity, need for cognition, or interest in complex systems.
- H3: The determining variable is technical training or STEM background, not general cognitive ability.

- H4: Preferences reflect cultural, identity, or socialization factors, independent of cognitive performance.

The H2–H4 numbering is local to this example: it labels competing causal explanations, not alternative hypotheses in the sense of the $H_0 \times H_1$ test. The canonical pair remains the usual one — H_0 is the null hypothesis and H_1 the alternative the study seeks to support.

A.2.3 Operational definitions

Cognitive performance.

We avoid reducing the concept of intelligence to a single score. Measurable proxies are proposed:

- abstract-reasoning test (for example, progressive matrices);
- cognitive-reflection measure (Cognitive Reflection Test, CRT, or equivalent);
- brief verbal-ability test;
- education and field of training as control variables.

Preference for *hard Sci-fi*.

Preference should be assessed through multiple indicators:

- self-report scales of preferred genres;
- forced-choice tasks between synopses with different degrees of scientific consistency;
- behavioral measures (retention, season completion, abandonment);
- scale of tolerance for physical or narrative inconsistencies.

A.2.4 Testable predictions

If H_1 is true, one expects to observe:

1. a positive correlation between cognitive proxies and declared preference for *Hard Sci-Fi*;

2. greater retention in narratives of high cognitive demand among individuals with higher cognitive performance;
3. partial persistence of the effect after controlling for age, sex, education, income, media consumption, and personality traits.

If mediating hypotheses are more appropriate, one expects that:

1. the effect of cognitive performance decreases substantially when mediators such as need for cognition are included;
2. there is a significant interaction between cognitive performance and technical training.

A.2.5 Research plan

A program composed of three complementary studies is proposed.

A.2.5.1 Study 1: correlational survey

Sample.

Broad and heterogeneous sample, with control of selection bias.

Instruments.

- brief cognitive tests;
- genre-preference scales;
- measures of openness to experience (distinguishing aesthetic and intellectual aspects);
- need for cognition and curiosity;
- sociodemographic variables.

Analysis.

Simple correlation, multiple regression, mediation models, and interaction tests.

A.2.5.2 Study 2: controlled experiment

Development of narratives equivalent in characters, conflict, and duration, varying only the degree of scientific consistency.

Measures.

- preferred choice;
- exposure time;
- objective comprehension;
- intention to continue.

Analysis.

Logistic models and hierarchical regressions to estimate the probability of choosing the more technically consistent version as a function of cognitive performance.

A.2.5.3 Study 3: ecological retention analysis

Collection of real behavioral data (retention, abandonment, completion) on platforms or through a voluntary digital diary, integrated with cognitive measures in a subsample.

Analysis.

Survival models (time to abandonment) and segmentation by cognitive profiles.

A.2.6 Implications for ROI and segmentation

If the association is confirmed, possible implications include:

- more precise audience segmentation;
- targeted marketing strategies;
- adjustment of audience expectations for specific niches;
- reassessment of success metrics, prioritizing retention and long-term value.

If the hypothesis is not confirmed, the results would still be informative, suggesting that factors such as marketing, competition, catalog saturation, or aesthetic preferences independent of cognition better explain commercial performance.

A.2.7 Methodological precautions

- Correlation does not imply causation.
- Need for rigorous control of confounders.
- Explicit and graded definition of *Hard Sci-Fi*.
- Avoid hierarchical or evaluative interpretations of the results.
- Transparency, pre-registration, and independent replication.

A.2.8 Methodological conclusion

The hypothesis of a correlation between cognitive performance and preference for *Hard Sci-Fi* constitutes an apt example of transforming an intuition into an empirically testable proposition. The disciplined application of the scientific method allows such a hypothesis to be confirmed, refined, or refuted on the basis of data, avoiding hasty conclusions and promoting well-grounded analysis.

A.3 Religious dogmatism as a cognitive obstacle and the separation between faith and science in John Locke

*Faith and reason are the two wings on which the human
spirit rises to the contemplation of truth.*

*Without the balance of both, flight is impossible:
with faith alone, one spins in circles under fanaticism;
with reason alone, one plunges into the cold materialism
that disregards ethics and the meaning of life.*

— Inspired by JOHN PAUL II, *Encyclical Fides et Ratio*

John Locke argued that religious dogmatism is not merely an undesirable moral trait, but a **cognitive and political error** that compromises rational inquiry and social coexistence. In the *Essay Concerning Human Understanding* (LOCKE, 1690), especially in Book IV, Locke devoted a chapter to the concept of *enthusiasm*, understood as the

acceptance of beliefs without empirical basis, sustained solely by the subjective conviction of a supposed inner revelation. For him, unexamined certainty undermines reason, prevents critical dialogue, and fosters intellectual intolerance.

Locke classified this “vice” along three main axes. First, **religious enthusiasm** (*fanaticism*) — belief assumed without sensory or argumentative evidence, based on the claim of direct divine inspiration. Second, **prejudice reinforced by blind authority** — the tendency to accept principles merely because they were transmitted by the family, the church, or prestigious institutions, without personal scrutiny. Third, the **political instrumentalization of faith** when church and State are mixed, as discussed in *A Letter Concerning Toleration* (LOCKE, 1689), a work in which Locke maintained that the church is a **voluntary** association, and that the use of state force to impose beliefs or “save souls” is logically incoherent, since coercion produces neither genuine conviction nor verifiable knowledge.

By rejecting the notion of innate ideas (*tabula rasa*² as the initial condition of the mind), Locke weakened the epistemological basis that had historically sustained unquestionable dogmas. For him, the **criterion of scientificity** must reside in propositions that derive from experience, are revisable by reason, and remain open to challenge — which is incompatible with any form of untestable certainty imposed as natural truth.

This Lockean distinction has direct implications for the contemporary scientific method: science must deal only with natural phenomena accessible to the senses and to intersubjective verification, remaining **non-dogmatic** on principle. Faith and religion, although legitimate as free societies and as individual experiences, cannot function as a criterion of scientific demarcation nor as epistemic authority in explanations about the physical or biological world when they are not amenable to empirical test or refutation.

In Box 2, the Lockean contrast is synthesized into comparative axes useful for critical demarcation.

Thus, Locke’s struggle against dogmatism is better described as an **epistemological antidote**: a commitment to experience, to critical reason, and to the functional separation between institutions of belief and institutions of natural inquiry. This principle does not eliminate faith from the social space, but prevents it from con-

² The Latin expression *tabula rasa* literally means “blank slate”. The metaphor comes from the ancient wax-covered tablets used for writing: by smoothing the wax, the surface became smooth again and ready for new inscriptions. John Locke used this image to defend the idea that the human mind is not born with ready-made or innate ideas, but begins as a blank page that is gradually “written” by sensory experiences and by learning over the course of life.

taminating the demarcation of what is considered science, ensuring that scientific activity preserves its most essential attribute: **permanent openness to revision and refutation**.

2. Axes of Locke's struggle against dogmatism and their relevance to science

- **Origin of knowledge:** derives from sensory experience, not from infallible authorities.
- **Reason and dialogue:** beliefs and theories must accept logical criticism and independent verification.
- **Institutional voluntariness:** churches are free associations; States regulate public order, not metaphysical convictions.
- **Demarcation:** only testable and revisable propositions may integrate the formal body of science.

A.4 Philosophical currents on the scientific method

A.4.1 Inductivism

Inductivism is a traditional conception according to which scientific knowledge derives directly from the careful and systematic observation of phenomena. The central idea is that, from a large number of particular observations, one can generalize universal laws. For example, by observing repeatedly that metals expand when heated, we could conclude that all metals expand with heat. Although intuitive and attractive, this model presents serious limitations: there is no secure logical criterion justifying the passage from the particular to the universal, as Hume had already pointed out (HUME, 1748). This fragility was extensively taken up again by modern philosophers, such as Popper and Chalmers (1999).

A classic example of this problem is the belief that “all swans are white”. From countless observations of white swans, inductivism could propose this generalization as a universal law. However, the simple discovery of a single black swan would suffice to falsify the hypothesis, showing the fragility of induction and the strength of the Popperian criterion of falsifiability.

A.4.2 Popper's falsificationism

Karl Popper proposed an alternative to inductivism, arguing that science does not progress through successive confirmations, but through the attempt to refute bold

hypotheses. For him, a scientific theory must be formulated so that it can be tested and, in principle, falsified.

3. Science as Clay Sculpture

Science is a clay sculpture under continuous modeling: experimental corroboration adds mass and gives shape to hypotheses, while falsificationism removes excess, corrects imperfections, and delimits the contours of the real by eliminating everything that does not correspond to reality. Throughout this process, new instruments, new methods, and new languages function as the sculptor's tools, allowing more precise cuts, bolder additions, and details previously impossible, expanding not only the work, but also the very capacity to sculpt it.

Scientific progress would then occur through the elimination of false theories, with those that resist criticism provisionally remaining (POPPER, 1935; POPPER, 1963). Thus, science is understood as a dynamic process of conjectures and refutations.

The Popperian dynamic of constructing and correcting theories is metaphorically synthesized in Box 3, which compares scientific progress to a continuous work of modeling and removing errors.

Box 4 shows some classic examples of Popperian falsificationism.

4. Examples of Popperian Falsificationism

- **Universal law:** "All swans are white."
Refutation: The observation of a single black swan falsifies the law.
- **Newtonian mechanics:** Prediction of planetary motion according to Newton's laws.
Empirical limitation: The anomaly in Mercury's orbit could not be explained by the Newtonian model, being later described by Einstein's general relativity.
- **Theory of spontaneous generation:** Life arises from inert matter under ordinary conditions.
Refutation: Pasteur's experiments (1861) showed that microorganisms do not appear without external contamination.
- **Ptolemy's model:** The Earth at the center of the universe (geocentrism).
Historical supersession: The observations of Copernicus, Galileo, and Kepler produced predictions incompatible with geocentrism, consolidating the heliocentric model.
- **Classical physics:** The speed of light depends on the motion of the observer.

continued →

4. Examples of Popperian Falsificationism (*continued*)

Refutation: The Michelson–Morley experiment (1887) showed that the speed of light is constant, paving the way for relativity.

- **Phlogiston theory:** Combustion occurs through the release of a substance called “phlogiston”.

Refutation: Lavoisier’s experiments (1770–1780) showed that combustion involves the consumption of oxygen, overturning the theory.

Beyond these historical examples, falsificationism is not an exceptional event but a permanent mechanism of scientific practice itself, in which theories and models are constantly subjected to empirical and conceptual criticism.

A.4.3 Kuhn and scientific revolutions

Thomas Kuhn brought a historical and sociological perspective to the philosophy of science. In *The Structure of Scientific Revolutions*, he argues that science operates, most of the time, in a regime of “normal science”, solving problems within an accepted paradigm. However, periods of crisis can lead to scientific revolutions, in which one paradigm is replaced by another incompatible with the previous one (KUHN, 1962). The notion of incommensurability between paradigms shows that scientific change is not merely a logical process, but also involves historical and social factors.

A.4.4 Lakatos and research programs

Imre Lakatos sought to combine Popper’s critical view with the historical dimension highlighted by Kuhn. His proposal of *research programs* distinguishes between a “hard core” of fundamental hypotheses, which scientists seek to protect, and a “protective belt” of auxiliary hypotheses, which can be modified in the face of anomalies. A research program is considered progressive when it manages to predict new phenomena and degenerative when it merely accommodates already known facts (LAKATOS, 1978). This conception seeks to preserve the rationality of science without ignoring its historical development.

A.4.5 Feyerabend and methodological anarchism

Paul Feyerabend took criticism to a radical level by defending what he called “methodological anarchism”. According to him, there are no universal rules governing sci-

entific practice; any attempt to fix a single method would be restrictive and even harmful to the advancement of knowledge. In his famous work *Against Method*, Feyerabend argues that, throughout history, scientists frequently broke accepted methodological rules and that, often, progress was possible precisely because of these ruptures (FEYERABEND, 1975). His motto “anything goes” expresses the conviction that the diversity of approaches is a strength of science.

Table 1 – Comparison of different conceptions of science

Current	Criterion of scientificity	Strengths	Limitations / Criticisms
Inductivism	Generalization from repeated observation.	Emphasis on sensory experience; intuitive model accessible to beginners.	Problem of induction (Hume); absence of logical justification for the passage from the particular to the universal.
Popper (Falsificationism)	Testability and falsifiability of hypotheses.	Critical and dynamic method; promotes bold and rigorously testable theories.	Not all science is formulated in falsifiable terms; in some fields, demarcation is less clear.
Kuhn (Paradigms and Revolutions)	Normal science governed by a paradigm; changes occur in scientific revolutions.	Historical and sociological emphasis; explains deep ruptures in knowledge.	The concept of incommensurability hinders comparison between paradigms; accused of relativism.
Lakatos (Research Programs)	Hard core protected by auxiliary hypotheses; evaluation by the progressiveness of the program.	Integrates historical dimension with rationality; offers an intermediate criterion between Popper and Kuhn.	Clearly distinguishing progressive from degenerative programs can be subjective.
Feyerabend (Methodological Anarchism)	Absence of fixed rules; “anything goes” as a principle.	Values the plurality of methods and scientific creativity; shows that transgressions can generate advances.	Considered excessively relativist; threatens to dissolve the boundaries between science and non-science.

A.5 The classical scientific method

The scientific method is frequently represented as a sequence of steps that allow observations to be transformed into theories, by means of testable hypotheses. In Figure 1, a didactic flowchart is shown depicting how these steps are articulated.

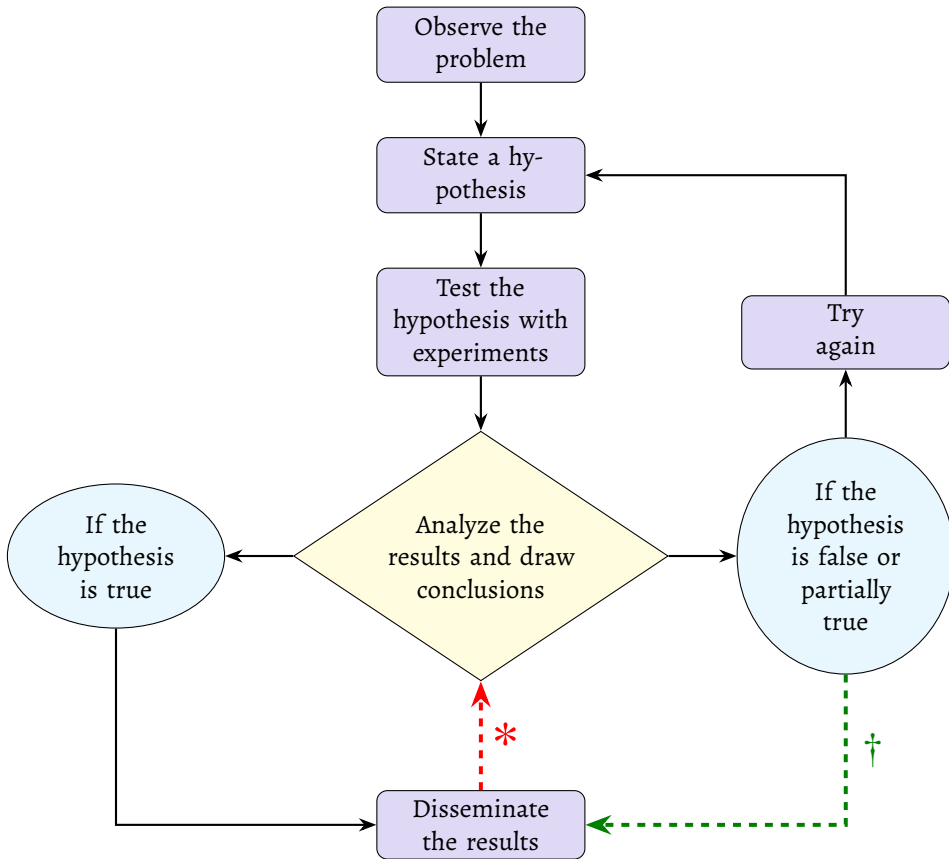


Figure 1 – Flowchart representing the steps of the classical scientific method. A theory is constantly tested and may be falsified at any moment or not (*). Often, false hypotheses are not disseminated, but they should be (†).



For a hypothesis to be considered scientific, it must be testable and, in principle, falsifiable. This process of continuous testing generates what we call a **theory**: an organized body of explanations that has survived criticism and attempts at refutation.

A theory is constantly tested and may be falsified at any moment. For this reason, it is often said that science is made of “provisional truths”: even if a theory is never

falsified, there will always exist the possibility that it may come to be falsified in the future.

It is important to stress that, although the flowchart presents the scientific method as a linear sequence of stages — observation, formulation of hypotheses, experimentation, analysis, and dissemination —, in practice the scientific process is more dynamic, nonlinear, and subject to constant revision. Historical, social, and cultural factors also exert influence on the way science is actually produced.

A.6 The position of this book in the flowchart of the scientific method

As we have seen, the flowchart of the scientific method represents the path that goes from observation to the consolidation of theories. This book situates itself in a specific way within that process: at the point “**State a hypothesis**”.

It is precisely at this moment that so-called **protoscience** is found: the initial stage in which ideas, models, and conjectures have not yet been exhaustively tested, but have the potential to guide future research.

All the effort developed here can be understood as an exercise in *protoscience*: the moment of raising well-grounded hypotheses that are, however, not yet experimentally validated. These hypotheses are presented, discussed, and related to already established knowledge, with the clear awareness that they are conjectures awaiting future testing.

Appendix B presents a more detailed discussion of the different forms of hypotheses.

Thus, this book deliberately occupies the initial phase of the scientific method, in which creativity, imagination, and the formulation of bold models take on a central role in the construction of knowledge.

A.7 Reputation in science

Reputation is a fundamental intangible asset in the scientific milieu. It functions as a *symbolic capital* that lends credibility to the researcher, influences collaborations, funding decisions, and even the acceptance of ideas at conferences and in journals.

A.7.1 Main sources of reputation

- **Academic and professional curriculum:** publications in high-impact journals; highly cited articles (h-index, impact factor); experience at renowned centers; invited lectures at international conferences.
- **Consistent scientific output:** original and relevant contributions; continuity of a solid line of research; opening new fields of investigation or solving important problems.
- **Recognition by peers:** positive evaluations in reviews; awards and distinctions; invitations to editorial boards; collaborations in international networks.
- **Integrity and ethical stance:** respect for ethical norms (avoiding plagiarism, data fabrication, and conflicts of interest); transparency and reproducibility; responsibility toward students and collaborators.
- **Social and scientific impact:** practical applications of discoveries; contribution to public policy and innovation; science communication and responsible participation in debates.

A.8 The Sagan effect and the institutional contradictions of science communication

Most research is funded by taxpayers, and they have the right to know what is being done with their money.

— AINISSA RAMIREZ, (RAMIREZ, 2020)

The so-called *Sagan Effect* designates the prejudice according to which scientists who are highly visible in the media would, for that very reason, be less serious or less productive researchers. The term refers to the astronomer Carl Sagan, whose extensive activity in science communication — through television series, books for the general public, and newspaper interviews — ended up earning him a certain distrust among academic colleagues. Despite his solid scientific credentials, Sagan was passed over for positions and even had his nomination to the *National Academy of Sciences* rejected in 1992, an episode that came to symbolize the stigma against popular scientists.³

The Sagan Effect manifests itself as a kind of symbolic sanction. As a researcher begins to speak to the public, write accessible books, or appear in the media, suspi-

³ On the historical context and the reception of Carl Sagan's work, see Martinez-Conde (2016) and Marsh (2019).

cion grows among some peers that he would be “self-promoting” or “abandoning scientific rigor”. This perception is reinforced by an academic culture that values measurable productivity — articles, citations, and bibliometric indices — more than the ability to translate scientific knowledge for society.

Empirical studies, however, show that this belief is unfounded. Research on the profile of scientists engaged in communication indicates that they tend to publish as much as their less visible colleagues and, in some cases, even more.⁴ The harm, therefore, does not occur in scientific output, but in academic reputation — that is, in the way the community interprets the scientist’s public exposure.

A.8.1 Informal sanctions and institutional contradictions

Recent qualitative research indicates that science communication is still permeated by institutional contradictions. Universities and funding agencies formally encourage public engagement, but maintain evaluation systems that scarcely recognize this type of activity. In practice, many researchers feel discouraged from participating in communication activities, fearing negative consequences for their careers.

In a study conducted by Calice et al. (2022), professors at a North American university reported that, although the institution proclaims support for outreach, the system of career progression continues to privilege journal publications and conventional metrics. The authors describe a “culture of dissonance”, in which the discourse valuing public engagement finds no support in the internal reward structures.

Similarly, Ho et al. (2020) observed among researchers in Singapore a tension between macro-institutional incentive — coming from funding bodies and public policy — and the meso-institutional reality, marked by administrative constraints and fear of informal penalties. Scientists recognize the social value of public communication, but report a lack of time, scarcity of resources, and fear of internal criticism.

This paradox is not new. As early as 1997, Gascoigne and Metcalfe (1997) showed that many Australian scientists found little institutional support — when not indifference — in approaching the media. In a later study, Jacobson, Butterill and Goering (2004) concluded that, even when universities claim to value “knowledge transfer”, the real incentives remain restricted to academic publication, reinforcing the gap between science and society.

⁴ See the analyses gathered in Martinez-Conde (2016), which reviews data on the productivity and public engagement of researchers from different fields.

Institutional research, such as that of Entradas and Bauer (2017), indicates that the degree of engagement of a research center depends strongly on organizational factors: the existence of specialized personnel, financial support, and explicit recognition of communication activities. When these elements are lacking, outreach becomes the voluntary initiative of a few motivated individuals, frequently overburdened. A similar situation was observed by Koivumäki and Wilkinson (2020) in Finland, where performance and public-visibility policies create conflicts of identity and inconsistent messages about the value of science communication.

The largest survey conducted on the subject confirms the mechanism and adds its converse. In a national survey of 8,533 Chinese scientists, Chen, Zhang and Jin (2023) found that negative experiences with public exposure feed negative personal norms, and that it is these — not the exposure itself — that inhibit the intention to take part in outreach. The same study shows that positive social norms and reward systems mitigate the effect, which shifts the question from the researcher's individual temperament to the signals the institution sends.

In sum, the so-called “informal sanctions” — distrust, lack of recognition, and absence of structural incentives — act as cultural barriers to the full legitimation of science communication. As long as communicating science to the public continues to be seen as a peripheral or risky activity, the Sagan Effect will persist as a reflection of an unresolved tension between the ideal of open science and the traditional mechanisms of academic prestige.

A.8.2 Envy, symbolic distinction, and the rhetoric of secrecy

Beyond institutional barriers, there are psychological and sociological factors that perpetuate the Sagan Effect. Part of some researchers' resistance to outreach seems to arise from a feeling of intellectual envy: the inability to translate complex ideas clearly can lead to the devaluation of those who do it successfully. Instead of recognizing their own didactic limitations, certain academics react by attacking communicators, labeling them “superficial” or “sensationalist”.

This attitude also reflects a form of symbolic distinction, in the Bourdieusian sense (BOURDIEU, 1991, p. 60): the maintenance of a hermetic and inaccessible language serves to preserve prestige and the monopoly of knowledge — and, for Bourdieu, this effect does not depend on intent, arising even when none of those involved means it. In many contexts, opacity functions as a currency of power — a way of making knowledge restricted, generating dependence and reinforcing

hierarchies. There are, therefore, those who create difficulties in order to “sell conveniences”, writing cabalistic or excessively technical texts not merely for conceptual precision, but as a strategy of symbolic self-protection.

These practices, by hiding knowledge behind a barrier of jargon, contribute to perpetuating the distance between science and society. True scientific didactics — that which clarifies without oversimplifying — requires intellectual courage, for it makes visible both the content and the researcher’s own limitations. In this sense, the Sagan Effect is not merely a problem of individual reputation, but a symptom of an academic culture that still confuses complexity with obscurity, and erudition with inaccessibility.

A.8.3 The most prestigious scientific awards

Among the main distinctions that consecrate a scientist’s reputation, the following stand out:

- **Nobel Prize** — awarded in Physics, Chemistry, Medicine, Literature, and Peace, it is the most widely recognized in the world.⁵
- **Fields Medal** — considered the “Nobel of Mathematics”, awarded every four years to researchers aged 40 or under.
- **Abel Prize** — conferred by the Norwegian Academy of Science and Letters in Mathematics.
- **Lasker Award** — focused on medical discoveries, often anticipating Nobel laureates in Medicine.
- **Breakthrough Prize** — created by technology investors, it recognizes advances in Physics, Life Sciences, and Mathematics.
- **Copley Medal** — one of the oldest awards in modern science, conferred by the Royal Society (United Kingdom).
- **Wolf Prize** — an international award in Sciences and Arts, with great academic prestige.
- **National awards** — distinctions of each country, which also impact local reputation.

These awards function as a validation of one’s career, often seen as the crowning of decades of research.

⁵ The Economics prize is not a Nobel: it is the *Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel*, established in 1968 by Sweden’s central bank on its 300th anniversary, and therefore outside the 1895 will. It is awarded at the same ceremony, which feeds the confusion.

A.8.4 The IgNobel prize

Among the awards that mark scientific reputation, there is a special distinction: the **IgNobel Prize**.

Created in 1991 by Marc Abrahams and organized by the magazine *Annals of Improbable Research*, it is awarded annually in a ceremony that for many years was held at Harvard University, with the participation even of Nobel laureates. The prizes honor achievements that “*make people laugh, then think*” (Improbable Research, 2026). The idea is to recognize real research, published in scientific journals, dealing with unusual, surprising, or apparently trivial phenomena that nonetheless invite reflection.

Why it is not dishonorable

- **Visibility:** it arouses public and media interest, bringing attention to little-explored areas.
- **Stimulus to critical thinking:** it highlights curious phenomena that might be forgotten without such prominence.
- **Recognition of creativity:** it values original and unexpected approaches.
- **Notable example:** the physicist Andre Geim received the IgNobel in 2000 for studying how frogs can levitate in magnetic fields, and ten years later received the Nobel Prize for his discoveries about graphene.

Possible negative perceptions

- In conservative contexts, the prize may be seen as mockery rather than recognition.
- The lay public may mistakenly interpret it as “worthless science”.
- The scientific impact of the awarded research varies: some bring relevant applications, others remain curiosities.

A typical example for an IgNobel: rotating universe and the cosmological principle

In modern astronomy, it is considered absurd to claim that the Earth is flat, that the Sun revolves around the Earth, and that all stars and planets describe orbits around our planet. These conceptions have been superseded since the advent of the

Copernican revolution, consolidated by Galileo's observations and the formulation of Kepler's laws (KUHN, 1957).

The recent hypothesis of a **rotating universe**, proposed by Szigeti et al. (2025), can be interpreted as a kind of *sophisticated reissue* of these archaic ideas: it is not a matter of placing the Earth at the center of the cosmos, but of admitting that the universe as a whole would rotate around a privileged axis. Just as geocentrism broke symmetry by postulating an arbitrary center, the hypothesis of global rotation breaks the **cosmological principle**, which assumes large-scale homogeneity and isotropy.

Although the mathematical formulation is ingenious, and was published in a prestigious journal (*Monthly Notices of the Royal Astronomical Society*), the idea that the entire universe would execute an extremely slow rotation (on the order of one turn every $\sim 3 \times 10^{12}$ years) contradicts both the simplicity of the Friedmann–Lemaître–Robertson–Walker model and the evidence of isotropy in the cosmic background radiation.

Thus, although useful as a theoretical exercise to test the limits of the Standard Model, such a hypothesis is closer to an **IgNobel Prize in Physics** than to a plausible explanation for the so-called Hubble tension⁶.

Example of an IgNobel awarded: *Boquila trifoliolata*, leaf mimicry, and open questions

The climbing vine *Boquila trifoliolata* presents the remarkable ability to copy the leaf morphology of neighboring plants — including artificial supports (such as plastic leaves) (WHITE; YAMASHITA, 2022); the extension to drawn two-dimensional patterns, without contact, is the hypothesis explored below. This phenomenon arouses great interest among scientists and laypeople, for it raises two central questions: (i) **how** a plant without eyes “perceives” the real or artificial leaves around it; and (ii) **how** the *genotype* is regulated so that, responsively, the leaf *morphology*, a *phenotype*, is modified within a single individual.

IgNobel 2024 (Botany).

⁶ The idea of a rotating universe had already been considered by Kurt Gödel in 1949 (GÖDEL, 1949), in an exact solution of Einstein's equations describing a rotating cosmos permeated by matter. However, that model implied the existence of *closed timelike curves*, opening the theoretical possibility of time travel — a result regarded as physically unacceptable. The current proposal by Szigeti et al. differs in postulating an extremely slow rotation, seeking to avoid such paradoxes, but it remains subject to similar criticisms regarding the breaking of isotropy and the observational difficulties.

In 2024, the IgNobel Prize in Botany was awarded to **Jacob White** (USA) and **Felipe Yamashita** (Brazilian, University of Bonn) for showing that real plants can mimic the leaves of *plastic* plants. The awarded work was published as “*Boquila trifoliolata* mimics leaves of an artificial plastic host plant” in the journal *Plant Signaling & Behavior* (WHITE; YAMASHITA, 2022)⁷.

The phenomenon and the open questions

Boquila adjusts the shape, size, color, and margin pattern (smooth/serrated) of its leaves according to the host on which it grows; a single branch can change successively as it moves between distinct hosts, suggesting external perception coupled to developmental responses. Two questions guide the investigation: (i) **which** signal(s) from the host/immediate environment the plant captures; (ii) **how** these signals are transduced to reprogram leaf development.

Perception hypotheses (not mutually exclusive)

- **Volatile chemical signals:** volatile organic compounds emitted by the host trigger internal response cascades.
- **Microbial mediation:** shared endophytic microorganisms convey small RNAs or other signaling molecules.
- **Optical patterns:** differences in reflected/absorbed light (intensity, spectrum, polarization, spatial frequency) modulate growth through advanced *photodetection*.
- **Mechanical signals:** micro-stimuli of contact (texture, stiffness) act as accessory cues.

Conditional inference.

If, under controlled conditions, it is unequivocally demonstrated that *Boquila trifoliolata* mimics both plastic leaves and drawn two-dimensional patterns (without contact), the most parsimonious remaining explanation is a pathway of **optical perception** — detection of light patterns (intensity, visible/UV/*near*-IR spectrum, polarization, spatial frequency/“texture”) reflected by the target, and not “vision” in the animal sense.

Minimum conditions to sustain this inference: (i) elimination/absorption of volatiles (sealed chamber with appropriate flow and filters); (ii) control/sterilization of the microbiome (inert/sterilized materials and surfaces); (iii) exclusion of mechanical

⁷ Official list of winners: <<https://improbable.com/ig/winners/>>. Article: White & Yamashita (2022), *Plant Signaling & Behavior*, DOI: 10.1080/15592324.2021.1977530.

contact and directional air currents; (iv) chemically inert test materials (non-emissive plastics/paints) with optical patterns manipulated in isolation.

From genotype to phenotype (mechanistic sketch)

Still without a definitive mechanism, a *plausible pathway* is:

External signal $\Rightarrow$ transduction (receptors/2nd messengers) $\Rightarrow$
 $\Rightarrow$ hormonal modulation (auxin, cytokinin, jasmonates) $\Rightarrow$
 $\Rightarrow$ epigenetic/transcriptional regulation $\Rightarrow$
 $\Rightarrow$ leaf developmental programs $\Rightarrow$ mimetic phenotype.

If under strictly controlled conditions (filtered/inert air, controlled microbiome, absence of contact, and use of chemically inert paints and substrates) it is demonstrated that *drawn* two-dimensional patterns induce mimicry in *Boquila trifoliolata*, then the most parsimonious remaining pathway is an **optical perception** based on the detection of light patterns (intensity, contrast, spectrum, and spatial frequency), followed by photosensitive transduction and developmental reprogramming. Predictions: (i) strong dependence on edges/contrast and on a band of spatial frequencies; (ii) spectral action compatible with plant photoreceptors (blue/UV and red/far-red); (iii) drop of the effect with blur/distance/angle; (iv) isoluminant patterns lose efficacy, suggesting dominance of luminance over chromaticity; (v) unstructured noise of the same luminance does not induce mimicry. A minimal coherent mechanism is: photodetection $\Rightarrow$ light regulators (PIFs/HY5/COP1) $\Rightarrow$ auxin gradients (PIN/ARF) in meristems $\Rightarrow$ modulation of shape genes (KNOX/CUC/TCP/YABBY) $\Rightarrow$ adjustment of margin, lamina, and veins.

In this framework, the phenotypic plasticity of *Boquila* stems from a genetic repertoire capable of activating alternative “traits” (margin, lamina, veins, pigments) in leaf meristems, under the control of gene networks modulated by plant hormones and epigenetic marks.

Testable predictions (examples)

1. **Air chemistry:** manipulations of the mixture of host volatiles, in sealed chambers, should reproduce (or abolish) mimetic patterns in a dose-dependent manner.

2. **Optics:** artificial visual filters and patterns (control of intensity, polarization, spatial texture) should induce systematic morphological changes if the main cue is photodetection.
3. **Microbiome:** treatments that alter the endophytic community (selective antibiotics/reintroduced consortia) should modulate the intensity/stability of the mimicry.
4. **Transcriptome/epigenome:** leaves in a distinct mimetic regime should exhibit differential signatures of gene expression and of histone methylation/acetylation in key genes of leaf development.

Gaps and implications

There is still a lack of (i) unequivocal causal demonstrations identifying the primary signal(s); (ii) detailed mapping of the transduction pathways; and (iii) comparative *multi-omics* (transcriptome/epigenome) under controlled conditions. Elucidating the mechanism of *Boquila* matters not only for the ecology and evolution of plasticity, but also for plant biotechnology (engineering of leaf shapes and chemical/optical camouflage) and for the general understanding of how biological systems integrate multiple environmental cues into developmental decisions.

Comparison with the golden raspberry

Unlike the cinema's *Golden Raspberry*, which is a direct criticism of the “worst”, the IgNobel is not a punishment. It is a good-humored celebration of research that, even while seeming bizarre, is real, published, and methodologically sound.

There is no institutionalized award marking the “worst scientists” or the “worst research”, but other initiatives have approached this role, such as the *Pigasus Award* (from James Randi, against pseudoscience and charlatanism) (James Randi Educational Foundation, 2007) or the *Golden Fleece Award* (in the USA, from Senator William Proxmire, against what he considered wasteful public spending) (United States Supreme Court, 1979, p. 114).

Historical reflection

Curiously, many ideas today considered central to science were seen as absurd in their time. Planck, Einstein, Wegener, Barbara McClintock, and Semmelweis

could, in their day, have been targets of satirical prizes for their radical ideas, which only later proved correct. This highlights the thin line between pseudoscience and frontier science, distinguishable only with time.

A.8.5 Factors that destroy scientific reputation

Despite its slow construction, reputation can be quickly shaken. Among the main factors are:

- **Scientific fraud:** fabrication or manipulation of experimental data.
 - *Example:* Hwang Woo-suk (South Korea), who falsified data on stem-cell cloning.
 - *Example:* Jan Hendrik Schön (Bell Labs), who published fraudulent results on molecular transistors.
- **Plagiarism and self-plagiarism:** appropriation of others' texts or the slicing of a single piece of research into multiple articles (*salami science*).
- **Denialism and pseudoscience:** adherence to ideas without scientific basis, such as denying climate change or defending miracle cures.
- **Misuse of resources:** financial diversion or inappropriate use of research funds.
- **Bad faith in collaborations:** improper appropriation of authorship, unfair exclusion of collaborators.
- **Deliberate scientific irrelevance:** production of publications without impact, in low-quality (predatory) journals.
- **Problems of writing and norms:** disregard for standard usage, non-compliance with journal submission rules.

A.8.6 Emblematic cases

- **Andrew Wakefield** — published in 1998 a fraudulent study linking vaccines to autism — retracted by *The Lancet* in 2010 —, resulting in a drop in immunization rates and serious consequences for public health (WAKEFIELD et al., 1998; GODLEE; SMITH; MARCOVITCH, 2011; BURGESS; BURGESS; LEASK, 2006; SMITH et al., 2008).

- **Pitldown Man** — paleontological fraud (1912–1953) that presented fake fossils as proof of human evolution, tarnishing the credibility of British scientific institutions (WEINER; OAKLEY; CLARK, 1953).

A.9 Scientific publication

The scientific method does not end with the formulation of hypotheses or with the obtaining of experimental results. The final and essential stage is **scientific publication**, the moment at which the knowledge produced is communicated to the community and submitted to critical evaluation.

Without publication, knowledge remains restricted to the researcher and is not integrated into the *scientific body*. It is through publication that an individual result becomes part of the collective heritage of science.

The main vehicle of communication is the **scientific article**, published in specialized journals. For it to fulfill its function adequately, it is important that publication occur in journals of significant impact for the field, ensuring visibility and relevance. In addition, the article must undergo the process of **peer review**, in which independent specialists examine the methodological quality, the consistency of the results, and the clarity of the exposition.

Peer review, although subject to criticisms and limitations, constitutes the main collective quality-control mechanism of science. It ensures that results are critically analyzed before integrating the body of accepted knowledge. Only after this validation can discoveries influence new research, guide technological applications, and consolidate theories.

Box 5 shows the main precautions necessary for scientific writing.

5. Essential precautions in scientific writing

- **Impersonality and sobriety:** use the impersonal/passive voice (“*it was observed*”); avoid the first person, unless the journal permits it.
- **Evidence and verifiability (no guesswork):** every claim must be supported by (a) reproducible experimental proof; (b) reference to articles published in journals of recognized impact and peer review; or (c) clear logical-mathematical deduction.
- **Materials and Methods:** present in detail the materials, equipment, software, parameters, and procedures, so that the experiment can be *fully reproduced* anywhere in the world, within the specified conditions.

continued →

5. Essential precautions in scientific writing (*continued*)

- **Precision:** define terms, avoid ambiguities, and quantify whenever possible.
- **Objectivity:** focus on data and results; avoid evaluative adjectives and value judgments.
- **Verbal consistency:** adopt uniform verb tenses (methods and results in the past; established concepts in the present).
- **Terminological rigor:** employ official nomenclatures and standards (e.g., SI, IUPAC), avoiding colloquial synonyms.
- **Absence of Figures of Speech:** do not use figurative expressions or puns that could compromise the clarity and accuracy of the information⁸.
- **Adequate references:** cite *all* sources of data, methods, and code according to the required standard (ABNT, APA, Vancouver, etc.).
- **Logical structure:** organize into coherent sections (Introduction, Methods, Results, Discussion, Conclusion), following the IMRyD or IMRAD standard (Table 3).
- **Reproducibility and traceability:** record protocols, software versions, seeds, calibrations, and make data/code available when possible.
- **Universality:** avoid regionalisms; use SI units and international standards.
- **Clarity and concision:** short, direct sentences without unnecessary ornament.
- **Research ethics:** declare ethics-committee approvals (in research with humans or animals), informed consent, and conflicts of interest.
- **Data transparency:** whenever possible, make raw data, code, and protocols available in public repositories, favoring open science.
- **Originality and integrity:** avoid plagiarism, self-plagiarism, and the undue fragmentation of results into multiple publications (*salami slicing*).
- **Limitations of the study:** explicitly acknowledge methodological limitations and potential biases, rather than omitting them.
- **Coherence between text and illustrations:** tables, figures, and graphs must complement the text, never replace it, and must be self-explanatory.
- **Citation and style norms:** faithfully follow the journal's standard regarding citations, bibliography, abbreviations, mathematical notation, and graphic style.
- **Standard usage of the language:** employ the standard variety of the language in which the article is written (Portuguese, English, etc.), avoiding colloquialisms, slang, regionalisms, or informal constructions.
- **Peer review and language:** the text must be written in formal language, grammatically revised and, if in English, preferably reviewed by native speakers of the field.

⁸ Note: The criterion “Absence of Figures of Speech” is deliberately **not** followed in this work, beginning with its very title, *The Metaphor of the Bubbleland*, and with the recurrent use of analogies throughout

It is important to stress that articles published in scientific journals are not, in general, committed to didactics. Their main objective is the formal recording of knowledge, with a focus on academic rigor and compliance with the field's norms of scientific communication. Expository clarity is desirable, but secondary relative to methodological and argumentative precision.

On the other hand, the didactic function — making knowledge accessible, integrated, and teachable — is exercised by technical and scientific books, as well as by lectures, courses, video lessons, and various other forms of academic and scientific outreach. These resources, although fundamental for education and dissemination, do not by themselves constitute the primary record of science. Thus, there is a division of roles: articles ensure the formal advance of the scientific body, while books, lectures, and other didactic materials promote its diffusion and pedagogical consolidation.

A.10 Metrics of scientific evaluation

*To steal ideas from one person is
plagiarism;
to steal from many is research.*

— WILSON MIZNER (1876–1933; attribution not confirmed)⁹

The challenge of citation does not lie properly in citing or reproducing a text, but in doing so excessively or without due acknowledgment of the source work. Citations, however, do not constitute an affront to authors but, on the contrary, represent a valuable indication that their work has relevance and influence.

the chapters. Such a choice is consistent with the *protoscientific* status of the work, which presents itself in the context of discovery (Subsection B.1.2 of Appendix B) and, for this reason, would not be suitable for publication in formal scientific journals. The remaining criteria of the Box remain as a reference for formal scientific writing.

⁹ The attribution is traditional but undocumented in Mizner's lifetime. The earliest known occurrence is from November 1929, in a *U.S. Forestry Service* bulletin, credited to Wallace Notestein, professor of History at Yale: "If you copy from one book, that's plagiarism; if you copy from many books, that's research"; the *Yale Book of Quotations* (SHAPIRO, 2006) records attributions to Mizner only from the 1940s onward (Quote Investigator, 2010). An 1820 forerunner, from Charles Caleb Colton: "If we steal thoughts from the moderns, it will be cried down as plagiarism; if from the ancients, it will be cried up as erudition".

A.10.1 The impact factor

The impact factor (IF) is a metric created by Eugene Garfield in the 1960s (GARFIELD, 1972), which measures the average number of citations received in a year by articles published in the previous two years in a given journal. For example, if a journal has $IF = 4$, this means that, on average, each article published in it in the previous two years was cited four times in the year in question.

Limitations and criticisms

Although widely used, the impact factor is subject to several criticisms:

- **Dependence on the research field:** areas with high publication and citation rates (such as biomedicine) tend to have much higher IFs than areas with slower paces (such as mathematics).
- **Unequal distribution:** most citations are concentrated in a few articles, so that the IF does not represent the real average well (SEGLEN, 1997).
- **Misuse:** it is frequently used to evaluate individual articles or researchers, when its function is to describe journals.
- **Perverse incentives:** it may stimulate questionable editorial practices, such as prioritizing review articles, which attract more citations.

A.10.2 Other journal metrics

Besides the impact factor, alternative metrics exist:

- **CiteScore** (Elsevier): similar to the IF, but with a four-year window.
- **Eigenfactor** and **Article Influence Score**: consider the citation network, assigning greater weight to more central journals.
- **SCImago Journal Rank (SJR)**: weights citations by the relevance of the citing journal.

A.10.3 Researcher metrics

The evaluation of researchers' productivity and impact usually relies on different bibliometric indicators, among which the following stand out:

- ***h-index***: the number h of an author's articles that each received at least h citations (HIRSCH, 2005).
- ***i₁₀ index***: the number of articles with at least ten citations (a metric adopted by Google Scholar).
- **Alternative metrics (altmetrics)**: include indicators derived from academic social networks, such as the *ResearchGate Score*, or from scientific dissemination platforms, such as *Academia.edu* (PRIEM et al., 2010).

A.10.4 Metrics of specific articles

Besides journal citations, it is now common to measure the immediate impact of articles through:

- **Altmetrics**: track mentions on social networks, blogs, Wikipedia, and other media (PRIEM et al., 2010).
- **Downloads and views**: provide a measure of initial interest, although not of consolidated scientific impact.

A.10.5 Institutional metrics

Institutions and universities are evaluated by international rankings (Times Higher Education, QS, Shanghai Ranking), which combine the number of publications, citations, awards, and academic reputation. These rankings are influential, but also the target of criticism for their lack of transparency and their bias in favor of large research centers.

A.10.6 Critical synthesis

Metrics are useful instruments, but should not be seen as substitutes for qualitative peer evaluation. Each metric reflects only one dimension of scientific activity and must be interpreted with caution. Indiscriminate use can distort research priorities and harm the diversity of scientific work. In this spirit, initiatives such as the *San Francisco Declaration on Research Assessment* (DORA) advocate limiting the use of the impact factor and valuing multiple forms of evaluation (San Francisco Declaration on Research Assessment, 2013).

A.11 The peer-review process

Peer review is the central validation mechanism in modern science. It consists of the critical evaluation of a manuscript submitted to a journal or conference by independent specialists in the same research field. Its objective is to verify methodological soundness, the originality of the results, the clarity of the presentation, and the relevance to the scientific community.

A.11.1 Review models

There are different modalities of peer review:

- **Single-blind review:** the reviewers know the identity of the authors, but the authors do not know the identity of the reviewers.
- **Double-blind review:** neither authors nor reviewers know each other's identities, seeking to reduce biases.
- **Open review:** the identities of authors and reviewers are known, and in some cases the reports are published together with the article.

A.11.2 Strengths of the system

- It acts as a quality filter, preventing the publication of works with serious flaws.
- It favors the improvement of manuscripts, since reviewers suggest corrections and refinements.
- It maintains a standard of rigor that sustains the trust of the scientific community.

A.11.3 Criticisms and limitations

Despite its importance, peer review presents several limitations (SMITH, 2006; IOANNIDIS, 2005):

- **Subjectivity:** different reviewers may issue contradictory reports on the same article.
- **Biases:** favoring large centers, renowned authors, or certain lines of research.

- **Delay:** the process can be slow, delaying the dissemination of important discoveries.
- **Conservatism:** highly innovative or disruptive ideas may face greater resistance.
- **Lack of transparency:** in many cases, the reports are not available to the public.

A.11.4 Alternatives and reforms

In recent years, proposals have arisen to increase the transparency and efficiency of the process:

- **Pre-publication in open repositories** (such as *arXiv* and *bioRxiv*), allowing prior public scrutiny.
- **Post-publication review:** articles are published quickly and evaluated openly by the community.
- **Open peer review:** reports are signed and made available together with accepted articles.
- **Use of digital platforms:** collaborative tools and public comments that make the process more agile and accessible.

A.11.5 Synthesis

Peer review remains the main form of quality control in science, but it should not be seen as infallible. It functions as a mechanism of screening and refinement, but it needs to be complemented by open-science practices, independent replication, and continuous critical evaluation. Thus, scientific credibility results from the combination of review, transparency, and reproducibility.

A.12 Who pays for scientific publication

Scientific publication is not neutral only in terms of knowledge; it also involves economic models that affect access, circulation, and even the direction of research. Traditionally, the costs of publication were covered by subscriptions from libraries and institutions, which restricted access to those who could pay for the journals. In recent decades, however, the *Open Access* movement has transformed this scenario.

A.12.1 Traditional subscription models

In the classic model, scientific journals are financed by subscriptions paid by universities, research centers and, on a smaller scale, individual readers. This system creates barriers to access, especially in countries and institutions with few resources. In addition, large publishers concentrate market power, charging high prices for journal packages.

A.12.2 Open access

The open-access model holds that scientific knowledge, being the fruit of collective effort and often funded with public resources, must be freely available to all. In this system, editorial costs are transferred from readers to authors or their institutions, generally in the form of *Article Processing Charges* (APCs). There are different routes of open access:

- **Gold route (*gold OA*):** the article is made freely available on the journal's own website, normally upon payment of an APC.
- **Green route (*green OA*):** the author deposits a version of the article (preprint or postprint) in an institutional or thematic repository.
- **Diamond route (*diamond OA*):** the article is published in open access at no cost to authors or readers, being financed by institutions, scientific societies, or governments.

A.12.3 Criticisms and challenges

Although open access has significantly broadened the democratization of knowledge, it has brought new problems:

- **High APCs:** can be prohibitive for researchers in countries with few resources.
- **Predatory journals:** exploit the APC model by charging fees without providing adequate peer review.
- **Global inequality:** even in open access, researchers from countries of the global South face difficulties in publishing in prestigious journals.

A.12.4 Synthesis

The debate over who pays for scientific publication reveals that science is not merely a process of generating knowledge, but also an economic and political system. While the open-access movement advances in the name of equity and transparency, the tension persists between the ideal of free knowledge and the real costs of its production and dissemination.

It is important to stress that scientists, in general, are not paid for their publications. In open-access models, what frequently occurs is the opposite: the authors themselves or their institutions must bear the *Article Processing Charges* (APCs), which can reach thousands of dollars per article. This model, although broadening the availability of knowledge, transfers the costs to those who produce the research, creating additional barriers especially for groups and institutions with few resources.

A.13 The dodo metaphor: insular tameness and the future of cognition in the age of artificial intelligence

The tragic fate of the dodo (*Raphus cucullatus*), a bird endemic to the island of Mauritius and extinct at the end of the seventeenth century (ANGST et al., 2017), offers a powerful allegory for the cognitive and sociocultural transformations that humanity faces with the advent of Generative Artificial Intelligences, especially Large Language Models (LLMs). Evolving in an isolated ecosystem devoid of natural predators, the dodo underwent an adaptive process that resulted in the atrophy of its wings and the development of so-called “insular tameness” — a complete absence of instincts of fear or defense in the face of external threats. When the biological environment underwent an abrupt rupture with the arrival of human navigators and, above all, of the mammals they introduced — monkeys, deer, pigs, and rats —, the bird’s perfect specialization for a frictionless world became its fatal vulnerability.

In drawing a parallel with human intellectual development, the proliferation of AI tools acts analogously to the elimination of predators in the dodo’s habitat: it removes the “cognitive friction” necessary for the strengthening of mental faculties. If the human mind comes to systematically delegate its analytical, interpretive, and creative functions to algorithms, there is a risk of inducing a form of intellectual insular tameness, whose impacts can be categorized in education, in the short term, and in the long term.

A.13.1 Impacts on education: the end of “pedagogical friction”

The ecosystem where the loss of “intellectual defenses” manifests most immediately is the educational environment. Historically, effective learning has always depended on a degree of resistance. Before the spread of AIs, students had to perform manual processes: seeking primary sources, taking notes on texts, structuring arguments from scratch, and erring repeatedly in mathematical calculations.

This process of “doing it by hand” was precisely the mechanism by which knowledge was internalized. The educational value did not reside in the delivery of the final work, but in the cognitive effort required to construct it. With AI taking on the role of executor, we observe:

- **The Illusion of Knowledge:** The student who generates a flawless essay in seconds confuses the tool’s capacity with his own erudition. He consumes the final result, but does not build the neural bridges needed to reach it autonomously.
- **Atrophy of Academic Resilience:** By eliminating the natural frustration that accompanies learning complex disciplines, generative tools create students who, like the dodo, do not know how to deal with difficulties. At the slightest sign of complexity, the instinctive response becomes the outsourcing of reasoning to the machine.

A.13.2 Short-term impacts: the degradation of operational fluency

In the immediate horizon, beyond the classroom, the impacts of severe dependence on AI manifest themselves in the vulnerability of society’s informational fabric:

- **Cognitive Laziness and Dependence on Crutches:** The ease of obtaining ready-made answers and automated summaries reduces the individual’s disposition for continuous learning. AI ceases to be a means of amplification (a tool) and becomes a substitute for the brain.
- **Erosion of Critical Thinking and Skepticism:** Users accustomed to cognitive passivity tend to accept the models’ outputs as absolute truths. Without the “defensive instinct” of skepticism, the individual becomes easy prey to algorithmic hallucinations and disinformation, just as the terrestrial bird was easy prey for sailors.

A.13.3 Long-term impacts: cultural mutation and neurological configuration

Although one or two generations of technological use will not alter the human genetic load, the intergenerational impacts manifest themselves profoundly in collective cultural and neurological evolution:

- **Alteration in Developmental Neuroplasticity:** The human brain shapes its networks based on the demands of the environment. If future generations grow up delegating the solving of complex problems and linguistic synthesis to LLMs, the neural pathways linked to critical thinking and deep abstraction run the risk of not fully consolidating during childhood and youth.
- **Fossilization of Human Knowledge:** If cultural, academic, and scientific production comes to be predominantly fed back by AI-generated texts, knowledge runs the risk of stagnating on an imitative plateau, recycled from past data, losing the spark of human creative error.
- **The Sociocognitive Bifurcation:** The long-term scenario points to a drastic division. On one side, a majority whose minds have atrophied due to absolute comfort (the heirs of insular tameness). On the other, an elite that will maintain analytical rigor and use AI not as a crutch, but as a *cognitive exoskeleton* to reach innovations previously impossible.

A.14 The replacement of technical and scientific books by LLM AIs

*As he saw it, people would come to love their own oppression, to adore the technologies that would undo their capacities to think.*¹⁰

— NEIL POSTMAN (interpreting ALDOUS HUXLEY)

A.14.1 Change in student behavior

Students, who constituted one of the largest slices of the editorial market for technical and scientific books, have come to use AIs based on language models (LLMs) for

¹⁰ In the foreword to *Amusing Ourselves to Death*, Postman (1985, Foreword) comments on the vision of Huxley (1932) presented in *Brave New World*. The phrase describes Huxley's prediction that humanity would end up loving its own oppression and adoring the technologies that would destroy its capacity to think.

various reasons: instant access, reduced or zero cost, speed of answers directed at the specific question, and greater interactivity. As a consequence, the traditional demand for technical and scientific books tends to decrease drastically.

A.14.2 Impact on the publishing market

With the drop in student demand, publishers of technical and scientific books face a contraction in print runs and in institutional purchases (libraries, courses, undergraduate programs). The viability of printed or digital books as a primary means of technical learning is being compromised. The book ceases to be the dominant instrument of study and takes on a role more of reference or conceptual record.

A.14.3 Epistemological consequence

The replacement of the technical book by AI entails profound changes in the way of learning: extensive and structured reading gives way to pointwise and fragmented consultation. Although AIs can provide quick answers, there is a risk of losing the overall view, conceptual depth, and the construction of critical thinking. Thus, the format of technical knowledge runs the risk of becoming utilitarian and immediatist, rather than reflective and systematic.

A.14.4 The outsourcing of thought and progressive dumbing-down

The current phenomenon represents not merely a technological change, but a cognitive and cultural transformation. More and more people are outsourcing their own reasoning: they cease to formulate hypotheses, confront ideas, or develop critical thinking, blindly trusting the answers provided by algorithms. LLMs, by offering ready-made conclusions made seductive by textual fluency, produce an illusion of understanding. The individual feels informed without, in fact, understanding. Curiosity is replaced by the immediate satisfaction of the answer, and the learning process becomes passive.

This cognitive delegation configures a process of **collective dumbing-down**: artificial intelligence takes on the role of thinking, while the user limits himself to consuming the results. The dystopian scenario caricatured in the film *Judge* (2006) thus becomes a realistic metaphor for the present — a society that maintains the appearance of technological intelligence, but progressively loses its reflective and critical capacity.

A.14.5 The paradox of negative feedback

Although LLMs replace technical and scientific books as a means of access to knowledge, they **depend** on those books (and on other structured, reviewed, and published human content) as a source of training. If technical and scientific books cease to exist or produce less quality content, where will the AIs obtain reliable data? This paradox reveals a negative feedback loop:

- AI reduces the production and use of technical and scientific books;
- With less book production, the corpus of high-quality human data shrinks;
- LLMs trained predominantly on synthetic data or on the recycling of outputs from other AIs suffer from a phenomenon called “*model collapse*” — irreversible losses of diversity and fidelity to the original human data (SHUMAILOV et al., 2024);
- Without new and structured human sources, AIs degrade in quality and reliability, compromising the ecosystem of human and automated technical knowledge.

Therefore, keeping alive the production of technical and scientific books and reviewed human content is not merely relevant to the publishing market — it is essential to the very epistemological sustainability of AIs.

A.15 Complements to the scientific method

A.15.1 Expert opinion versus evidence

It is worth emphasizing that scientific methodology is not based on personal opinions, even when coming from renowned specialists. Scientific authority does not reside in individual prestige, but in the possibility of testing, replication, and falsification of propositions. Thus, even the opinions of specialists acquire scientific validity only when grounded in empirical evidence and amenable to independent verification.

A clear example is found in the medical field: the opinion of physicians, however qualified they may be, does not definitively validate a medication or vaccine. For that, a *randomized controlled trial* (RCT) is necessary, in which groups of patients are

compared under statistically controlled conditions. Only this type of evidence allows one to assert, with scientific rigor, the efficacy and safety of an intervention.

This distinction between blind authority and testable evidence had already been pointed out by Locke as a cognitive obstacle to rational knowledge (Section A.3).

A.15.2 Reproducibility and replicability

Scientific knowledge is consolidated only when it can be reproduced or replicated. Reproducibility refers to the capacity to repeat an experiment with the same data and methods, obtaining identical results. Replicability implies repeating the study under different but logically equivalent conditions, confirming the conclusions. The so-called “replication crisis”, especially notable in the biomedical and social sciences, highlights the importance of these criteria (Open Science Collaboration, 2015; IOANNIDIS, 2005).

A.15.3 Open science and open data

The open-science movement advocates the public availability of data, code, and experimental protocols. This sharing increases transparency, allows independent verification, and combats publication bias. Initiatives such as data repositories and preprints are contemporary examples of this advance.

A.15.4 Scientific models

Models are not copies of reality, but simplified representations that allow one to understand, predict, or simulate phenomena. The validity of a model is not measured by absolute fidelity, but by its usefulness, internal consistency, and capacity to generate testable predictions.

A.15.5 Error and uncertainty

Every measurement involves uncertainty. The use of statistical tools, confidence intervals, and significance tests does not eliminate error, but incorporates it in a controlled way into the interpretation of the results. The recognition and adequate treatment of uncertainty constitute one of the pillars of the scientific method.

A.15.6 Limits of the scientific method

Science is an enterprise limited to the natural and testable world. Questions of a supernatural, ethical, or aesthetic order cannot be resolved by scientific methods alone. Recognizing these limits does not weaken science; on the contrary, it strengthens its legitimacy as a specific mode of knowledge production.

A.15.7 Emerging paradigms and contemporary science

The development of digital technologies, computational simulations, artificial intelligence, and the analysis of large volumes of data (*big data*) expands the reach of the classical scientific method. These tools do not replace the fundamental stages of hypothesis, test, and falsification, but extend their application to new domains.

A.15.8 Confirmation bias in scientific research

“Facts are stubborn, but our minds are even more so.”

— ANONYMOUS

Confirmation bias is a cognitive tendency by which individuals preferentially seek, interpret, and value information that confirms their prior beliefs or hypotheses, ignoring or underestimating contrary evidence. Even in scientific practice, where neutrality and objectivity are presumed, this bias can significantly influence the way data are collected, analyzed, and interpreted.

According to Nickerson (1998), confirmation bias is ubiquitous and manifests itself in various forms of human reasoning, including scientific investigation. Among the most common manifestations of this bias are: (i) the unconscious selection of data that reinforce the researcher’s hypothesis; (ii) the biased interpretation of ambiguous results; and (iii) the omission or devaluation of evidence that contradicts the initial expectations.

In this context, statistics can, inadvertently, become a tool for reinforcing bias. The economist Ronald Coase ironically observed that “*if you torture the data long enough, it will confess to anything*”¹¹. In the context of confirmation bias, this maxim acquires a particular meaning: statistics is used to *torture the numbers until they confess exactly*

¹¹ Authorship of the phrase is generally attributed to Ronald H. Coase, a British economist and Nobel laureate, although there is no direct textual record in his works. The quotation commonly appears associated with the essay *How Should Economists Choose?* (1981), and is widely accepted in academic circles as an expression of his critical and ironic style.

what the researcher wishes to hear. The sophistication of quantitative methods does not eliminate the risk of biased interpretations — it only makes them harder to detect.

In the context of scientific communication, this phenomenon is also expressed through so-called *publication bias*, by which studies with positive results are more likely to be published than those presenting negative or null results. This tendency creates a distorted perception of the validity of certain hypotheses and compromises the reproducibility of science.

As Kuhn (1962) observed, during periods of “normal science”, the scientific community tends to operate within established paradigms, often disregarding anomalies that could lead to a revision of its fundamental theories. This collective behavior is an amplified expression of confirmation bias, since it reflects the community’s own resistance to questioning its presuppositions.

For Popper (1935), by contrast, the scientific method should oppose precisely this tendency, basing itself on the *falsifiability* of hypotheses: a scientific theory must be constructed so that it can be refuted by experience. When confirmation bias predominates, this principle is undermined, and scientific progress becomes slower and less critical.

A.15.8.1 Strategies for mitigating confirmation bias

Various approaches have been proposed to reduce the influence of confirmation bias in the scientific process. Among the most relevant, the following stand out:

- a) **Pre-registration of hypotheses:** defining and publicly registering the hypotheses, methods, and analysis criteria before data collection, avoiding subsequent manipulations.

Despite its methodological importance, the pre-registration of hypotheses is still a **rare practice** in most scientific fields. According to Nosek et al. (2018), its adoption has spread in a concentrated way — in experimental psychology, the behavioral sciences, and clinical trials, the latter by regulatory requirement — and remains incipient in most other fields. This scarcity stems, in part, from the lack of institutional incentives, the fear of premature exposure of ideas, and the persistence of a culture that values positive discoveries more than methodological transparency. From an epistemological standpoint, this gap allows the line between exploratory research and confirmatory testing to remain blurred — fertile ground for confirmation bias to act silently under the appearance of statistical rigor. *Pre-registration, therefore, is not merely a technical*

resource, but an epistemological defense against the subtle influence of confirmation bias.

- b) **Double-blind and randomized experiments:** preventing both the researcher and the participant from knowing which experimental condition is being applied, reducing unconscious interference.
- c) **Reproducibility and transparency:** publicly making available the datasets, code, and experimental protocols, allowing independent verification of the results.
- d) **Publication of negative results:** encouraging scientific journals and funding agencies to value studies that refute hypotheses, avoiding publication bias, which is discussed in Subsection A.18.1.
- e) **Meta-analyses and systematic reviews:** synthesizing multiple studies quantitatively, diluting the influence of isolated results and personal interpretations.
- f) **Critical scientific education:** promoting the training of researchers aware of their own cognitive biases, strengthening epistemological vigilance and critical thinking.

The recognition of confirmation bias and the conscious effort to mitigate it are, therefore, indispensable to the integrity and advancement of science. Only through constant epistemological vigilance is it possible to approach the ideal of objectivity that grounds the scientific enterprise.

A.16 Why does *The Metaphor of the Bubbleland* take the form of a book and not an article?

Throughout the history of science, some decisive conceptual advances were presented directly in books. Isaac Newton published the foundations of classical mechanics and universal gravitation in the *Philosophiae Naturalis Principia Mathematica* (1687), and not in short articles. Charles Darwin described natural selection and proposed a new unified vision for biology in *On the Origin of Species* (1859), again in book form. In these works, the aim was not merely to report an isolated experimental result, but to establish a new way of understanding nature, with consequences too broad to fit within a narrow scope.

In the twentieth century, this authorial function of the book practically disappeared within dominant scientific practice. Science institutionalized itself around the short scientific article, published in specialized journals, peer-reviewed, and indexed in bibliographic databases. This model became hegemonic after the Second World War, since it meets four central demands of contemporary science: (i) speed of publication and priority of discovery; (ii) methodological and formal standardization; (iii) quantification of academic productivity; and (iv) ease of bureaucratic evaluation of curricula and funding (KUHN, 1962). In this arrangement, books came to be seen above all as didactic materials, field reviews, or historical syntheses, and no longer as primary vehicles of original scientific proposition.

Even so, some ideas that attempt to reformulate entire conceptual domains continue to appear as books and not as articles. This occurs mainly when the proposal is too broad to be reduced to a single experiment, or too interdisciplinary to fit within the limits of a specific-field journal. Important examples of this type of work include:

- Ilya Prigogine, in *From Being to Becoming* (1980), in proposing a rereading of irreversibility, of thermodynamics far from equilibrium, and of physical time as a constructive element of reality, and not a mere parameter (PRIGOGINE, 1980).
- David Bohm, in works such as *Wholeness and the Implicate Order* (1980), in suggesting a physical ontology based on implicate/explicate order, which attempts to go beyond the purely instrumental reading of quantum mechanics and reinstate an unfragmented view of totality (BOHM, 1980).
- Stuart Kauffman, in *At Home in the Universe* (1995), in arguing that self-organization plays as fundamental a role as natural selection in the emergence of biological order, and in proposing fitness landscapes and laws of complexity that go beyond strict Darwinism (KAUFFMAN, 1995).
- Roger Penrose, in *The Emperor's New Mind* (1989), and later in *Shadows of the Mind* (1994), in arguing that the human mind cannot be fully reduced to algorithmic computation and in suggesting possible non-computational (and even quantum-gravitational) roles in consciousness (PENROSE, 1989; PENROSE, 1994). This is a thesis that crosses mathematics, fundamental physics, and neuroscience, impossible to present in traditional article form without structural loss.

- Stephen Wolfram, in *A New Kind of Science* (2002), in arguing that simple computational systems, such as cellular automata, constitute an alternative route for formulating laws of nature and generating complex physical behavior. The book is not merely a compilation: it attempts to inaugurate an entire research program with its own ontology and method (WOLFRAM, 2002).
- Lee Smolin, in *The Trouble with Physics* (2006), in making an internal critique of the institutional direction of high-energy theoretical physics, especially regarding supersymmetry and string theory, and in discussing the impact of academic power structures on the effective advancement of science (SMOLIN, 2006). Here, beyond the technical content, there is also a sociological critique of the scientific process.

These examples share a common trait: all of them go beyond the format of the scientific article because they are not merely reporting “a result”. They are attempting to propose a *research program*. In many cases, this program is still pre-paradigmatic, that is, it is at a phase in which there is not yet a consolidated community, nor consensual validation criteria, nor a single formal language. Thomas Kuhn describes this stage as a moment prior to “normal science”: it is when multiple approaches dispute the basic conceptual articulation of the field (KUHN, 1962). At this stage, the book fulfills the role of a foundational manifesto.

Moreover, several of these authors explicitly dealt with themes that cross disciplinary boundaries: thermodynamics and philosophy of time (Prigogine), quantum physics and ontology (Bohm), evolutionary biology and complexity theory (Kauffman), mathematical logic and consciousness (Penrose), computation and laws of nature (Wolfram), theoretical physics and sociology of science (Smolin). A conventional technical journal, devoted to a single domain, tends to reject such texts not necessarily out of empirical disagreement, but for lack of editorial framing. The book, therefore, becomes the only vessel capable of carrying the complete proposal without mutilating it.

This work situates itself in that same tradition. It does not seek merely to answer a circumscribed experimental question, but to propose a broad conceptual structure, with hypotheses about the functioning of physical and biological reality, with epistemological and methodological consequences. This proposal involves distinct levels (physical, biological, computational, cognitive) articulated together. This is not amenable to publication as a conventional article because:

1. The minimal unit here is not the “result”, but the *framework*. The objective is not to report a pointwise experiment, but to present a way of understanding phenomena that do not yet even possess a consensual formal language.
2. The argumentation is cumulative and narrative. Conceptual parts in Physics, Biology, Computation, and Philosophy need to be prepared and then connected to one another. Fragmenting this into isolated articles would lose precisely the integrating element that constitutes the central contribution.
3. Some hypotheses are still, deliberately, protoscience: they are proposals of mechanisms, ontologies, and interpretations that cannot yet be fully formalized in accepted equations or tested in the laboratory. At this stage, the book functions as a statement of intent for a future research program, not as a final report.

For these reasons, the book format adopted here is not merely a stylistic choice, but a structural necessity. It follows a rare, yet existing, lineage of works that attempt to inaugurate a new conceptual field before that field has become orthodox enough to fit into specialized journals.

A.16.1 Carl Sagan’s lessons: the baloney detection kit

But in introducing me simultaneously to skepticism and to wonder, they taught me the two uneasily cohabiting modes of thought that are central to the scientific method.

— CARL SAGAN, *The Demon-Haunted World*, preface

If the preceding sections established *what* distinguishes a scientific proposition from a dogmatic or pseudoscientific one, the practical question remains: *how* can a reader — researcher or not — put that distinction to work when facing a concrete claim. Carl Sagan devoted Chapter 12 of *The Demon-Haunted World* to that question, under the title “The Fine Art of Baloney Detection” (SAGAN, 1997).¹²

What Sagan proposes is a *toolkit* of skeptical thinking: a set of procedures that allows one both to build well-founded arguments and to identify and reject claims that are fallacious, fraudulent, or simply empty. The decisive point — and the reason

¹² *Baloney* colloquially designates “empty talk” or “nonsense”, which is why the instrument is commonly referred to as the *baloney detection kit*. The label should not be read too narrowly: the kit is aimed not only at deliberate falsehood, but also at sincere error, self-deception, and rhetorical imprecision.

the instrument belongs in this appendix — is that these tools are no methodological privilege of the laboratory: they are a condition of intellectual autonomy in the face of government propaganda, consumer fraud, and commercial pseudoscience.

Constructive tools

1. **Independent confirmation of the facts:** whenever possible, the data alleged must be verified by a source external to, and independent of, whoever presents them (see Subsection A.15.2).
2. **Encouragement of substantive debate:** to promote in-depth discussion of the evidence, involving specialists who genuinely know the different points of view in dispute — and not only the advocates of a single position.
3. **Distrust of arguments from authority:** in science there are no unquestionable authorities; there are, at most, experts. Authorities have erred in the past and will go on erring (see Section A.3, on prejudice reinforced by blind authority, and Subsection A.15.1).
4. **Multiple working hypotheses:** faced with a problem, to formulate every plausible explanation one can conceive and submit each of them to systematic attempts at refutation. The hypothesis that survives this “Darwinian selection” has a far greater probability of being correct than one that never had competitors (see the rival hypotheses H2–H4 of Subsection A.2.2).
5. **Detachment from one’s own hypothesis:** to avoid insisting on an explanation merely because it was the first to occur, or out of affective attachment to it. This is the direct antidote to the confirmation bias discussed in Subsection A.15.8.
6. **Quantification:** to seek numerical data and measurements whenever feasible, since what is vague and purely qualitative is far more vulnerable to error and manipulation (see Subsection A.2.3, on operational definitions, and Subsection A.15.5, on the treatment of uncertainty).
7. **Consistency of every link:** if the argument depends on a chain of reasoning, every link — including the initial premise — must work, and not merely most of them. An argumentative chain is no stronger than its weakest link.
8. **Occam’s razor:** between two hypotheses that account for the same data equally well, prefer the simpler one.¹³

¹³ This is a heuristic of choice, not a criterion of truth: simplicity guides a *provisional* preference, but does not settle empirical adequacy. Simple hypotheses can be false, and nature is under no obligation to be economical.

9. **Falsifiability:** to ask whether the hypothesis can, at least in principle, be tested and shown to be false. Propositions that offer no opportunity whatsoever for refutation have no real practical value (see Subsection A.4.2 and Section A.17).

Fallacies to recognize

Beyond the guidelines for sound analysis, the kit also indicates what *not* to do, cataloging the logical and rhetorical fallacies most frequently used to mislead. The principal ones are gathered in Box 6.

6. Frequent logical and rhetorical fallacies, according to Sagan

- **Ad hominem:** attacking whoever advances the argument, rather than the argument advanced.
- **Argument from authority:** validating an idea by the social, institutional, or power position of whoever states it.
- **Begging the question:** including in the premise the very conclusion one intends to demonstrate.
- **Observational selection** (“counting the hits and forgetting the misses”): recalling only the favorable cases and the coincidences, systematically ignoring the failures.
- **Excluded middle** (*false dichotomy*): forcing a choice between two extremes, suppressing the intermediate possibilities.
- **Confusion of correlation and causation:** supposing that, because two events occur together or are statistically correlated, one necessarily caused the other (see Subsection A.2.7).
- **Straw man:** caricaturing or distorting the opponent’s position in order to make it easier to attack.
- **Suppressed evidence, or half-truths:** presenting real data incompletely or out of context, concealing the unfavorable facts.

Reflexive application to this work

A work that declares itself protoscientific has a particular obligation to submit itself to the kit — on pain of claiming for itself the indulgence it denies to everyone else. Three consequences follow from this:

- **No hypothesis stands on the author’s authority.** That is precisely the function of the admissibility criteria **Cr1–Cr5** (Appendix B, Section B.2): to make the assessment public and auditable, rather than dependent on who signs it.

- **Rival hypotheses must be made explicit, not omitted.** The tool of multiple working hypotheses requires that the competing explanations — including the conventional ones — be presented to the reader, whose task is then to check whether the proposed alternative survives the attempt at refutation better.
- **Falsifiability as a condition of admission, not as an achievement.** As stated in Section A.6, this work stands at the stage of *stating hypotheses*. That does not exempt it from the ninth tool: a hypothesis not yet tested remains legitimate; an *untestable* hypothesis does not.

In this sense, Sagan's kit is not merely one more instrument presented in this appendix: it is the standard by which the rest of the book asks to be judged.

A.17 Teleological arguments and scientific status

Teleological arguments start from the idea that natural phenomena or biological structures exist as a function of a *purpose* or *finality*. In the classical Aristotelian formulation, this is the so-called *final cause*. Although they played a central role in ancient natural philosophy, these arguments are not accepted as scientific in modern epistemology, for well-established reasons.

A.17.1 Scientific limitations of teleology

According to Karl Popper's criterion of falsifiability, for a proposition to be considered scientific, it must be *testable* and *falsifiable*. A teleological argument does not meet this requirement, since it links phenomena to intentions or finalities that are not empirically observable. Whereas mechanistic hypotheses describe cause-and-effect relations, teleological arguments operate with the logic of purpose and effect, in which the purpose cannot be directly verified or refuted.

In modern physics, chemistry, and biology, teleological language is used only metaphorically or didactically. For example, it is common to say that “wings serve for flying”, but the scientific explanation is given in terms of selective pressures and adaptive advantages, without reference to a “conscious purpose” of nature.

A.17.2 Protoscientific value

Although not scientific in themselves, teleological arguments can have *protoscientific* value. This occurs when they function as heuristics for raising investigable hypothe-

ses. Historically, many discoveries arose from questions expressed in teleological language:

- “What would the purpose of the liver be?” motivated research on metabolism.
- “What are blood vessels for?” stimulated investigations on circulation.

In these cases, teleology was not maintained as an explanation, but served as initial motivation for experiments that resulted in mechanistic theories.

In this book, *The Metaphor of the Bubbleland*, some teleological arguments were used in the hope of motivating future experiments.

A.17.3 Risk of pseudoscience

If, instead of serving as a heuristic, teleology remains only at the level of untestable finalities, and is nonetheless presented as a scientific explanation, then it falls into the category of *pseudoscience*. Examples include versions of *intelligent design* that present themselves as science but do not meet the methodological criteria of testability and falsifiability.

A.17.4 Gradient of teleology

The epistemological status of teleology can be organized into three levels: philosophical/theological, protoscientific, and pseudoscientific, as shown in Table 2.

Table 2 – Teleology: Philosophical, Protoscientific, and Pseudoscientific

Criterion	Philosophical/ Theological	Protoscientific	Pseudoscientific
Definition	Explanations in terms of transcendent or metaphysical finality.	Initial inspiration for testable scientific hypotheses.	Teleological explanations presented as science, without testability.
Historical example	Aristotle: the eye exists in order to see.	Hypothesis about the function of the liver leading to physiological research.	Intelligent design presented as a scientific theory.
Testability	Not testable.	Can generate testable hypotheses.	Not testable.
Epistemological Status	Philosophy or theology.	Protoscientific (pre-scientific stage).	Pseudoscience.

A.17.5 Teleology in *The Metaphor of the Bubbleland*

In *The Metaphor of the Bubbleland*, elements of a teleological character were employed, but not in the classical Aristotelian sense of *final cause*, nor in the pseudoscientific sense. The recourse to teleology appears as a heuristic tool inspiring hypotheses, situating itself, therefore, in the field of the *protoscientific*.

Function and purpose of structures

In several passages, biological molecules such as proteins, DNA, and RNA are described as acting in an apparently *intentional* or *finalistic* manner, a hypothesis linked to the idea that such biomolecules could interact with parallel dimensions or realities. Although the formulation is teleological, the aim is not to attribute a metaphysical finality, but to open space for new investigable hypotheses.

Greater purpose of the metaphor

Bubbleland itself arises with the purpose of answering questions of the type “what is the point of considering parallel realities?”. The proposed answer is that this metaphorical resource can elucidate phenomena still unexplained in areas such as molecular biology and the study of consciousness. Thus, teleology appears as an initial research motivation, not as a conclusive explanation.

Epistemological value

The presence of teleological elements in Bubbleland must be understood within the protoscientific status:

- they function as inspiration for hypotheses,
- they guide the formulation of investigable questions,
- they do not intend to replace the scientific method with explanations based on finalities.

Synthesis

Therefore, Bubbleland uses teleology in a strategic and provisional way, as a resource for proposing new lines of investigation. This is a protoscientific use, which differs clearly from the pseudoscientific employment of untestable finalities presented as if they were science.

A.18 Gaps in the scientific literature: falsifications and propositions

Currently, the published scientific literature concentrates predominantly on positive results, that is, confirmed hypotheses or, at least, ones not refuted. This pattern creates two important voids: the absence of a systematic literature of falsified hypotheses and the lack of a formal space for protoscientific propositions.

A.18.1 Literature of falsified hypotheses

Frequently, unsustained hypotheses end up being shelved or appear only in little-highlighted sections of articles, such as “limitations” or “future work”. This phenomenon generates so-called *publication bias*, in which only positive results circulate, while negative ones are lost (IOANNIDIS, 2005).

If there were a literature dedicated to refuted hypotheses, researchers would not repeat work already shown to be flawed, saving time and resources. Moreover, science itself would become more robust, clearly showing what *does not work*. Some partial initiatives in this direction already exist, such as *negative-results journals*, although they are still little valued; the Journal of Negative Results in Biomedicine (2017), devoted to exactly that, ceased publication, with its archive still accessible.

A.18.1.1 Historical examples

The emphasis on positive results was not as pronounced in the past. Philosophers such as Popper (1935) highlighted the importance of *falsification* as a criterion of scientificity. For Popper, hypotheses that resist attempts at refutation become more solid; and refuted hypotheses must be equally valued, since they delimit the field of knowledge.

A.18.1.2 Contemporary examples

Today, there exist some journals and initiatives that seek to record negative or non-confirmatory results, such as the Journal of Articles in Support of the Null Hypothesis (s.d.). Open data platforms also offer space to archive experiments that did not confirm the initial hypotheses.

A.18.2 Protoscientific literature

Another void concerns the systematic proposition of hypotheses before the existence of robust experimental proof. Today, this space is partially occupied by opinion articles, perspectives, essays, or even preprints, all with editorial limitations that discourage the proposition of bold hypotheses.

A protoscientific literature would allow promising ideas to be recorded in a structured way, creating an open base of testable hypotheses that could be taken up and verified by other research groups. This type of publication would lend legitimacy to scientific creativity, encouraging the formulation and debate of hypotheses in a more transparent and organized manner.

A.18.2.1 Why are modern protoscientific books rare?

The scarcity of books labeled “protoscientific” in modern times is due to three main factors:

- **Acceleration of the Scientific Method:** Today, the interval between a speculative hypothesis and its testing in the laboratory or via computational simulation has drastically decreased. Many ideas that would be “protoscientific” are quickly validated or discarded in academic articles even before they become books.
- **Institutionalization of Science:** The peer-review system tends to absorb well-grounded speculations as “frontier theoretical physics” or “speculative biology”, rather than leaving them on the margin as protoscience.
- **Stigma:** The term “protoscientific” is frequently avoided by serious authors so as not to be confused with the esoteric or “quantum self-help” literature that dominates the shelves of speculative nonfiction.

A.18.2.2 Historical examples

The natural philosophy of the seventeenth and eighteenth centuries can be seen as a form of protoscientific literature. The works of Descartes (1637), Leibniz (1714), and Newton (1704), when they still speculated about invisible forces or the nature of light, had a strongly hypothetical and speculative character before receiving experimental validation.

A.18.2.3 Contemporary examples

Today, some digital platforms already partially fulfill this role. *arXiv* (arXiv, 1991) or *Zenodo* (European Organization For Nuclear Research; OpenAIRE, 2013), for example, allow the dissemination of ideas not yet peer-reviewed, functioning as an open space for hypotheses. Other initiatives, such as Hypothesis Project (2011), enable collaborative annotations on published articles, creating a layer of protoscientific propositions and discussions.

Although rare, it is possible to find books with a protoscientific approach that explore themes such as:

- Non-Local Consciousness: Attempts to explain consciousness via quantum mechanics (e.g., theories of Roger Penrose).
- Exobiology / Astrobiology: Speculations about non-carbon-based forms of life.
- Frontier Technologies: Theoretical proposals about warp-drive propulsion or sentient artificial intelligence.

A.18.3 Connection with this book

While the literature of falsified hypotheses would attack publication bias and enrich the map of knowledge, the protoscientific literature would open space for creativity, broadening the legitimacy of well-grounded and collaborative speculation within science. Historical and contemporary examples show that both the recording of refuted hypotheses and the systematic proposition of speculative hypotheses are fundamental to a more transparent and innovative scientific ecosystem.

This book can be understood as an exercise in that spirit: an attempt to record protoscientific hypotheses in a systematic and public way, contributing to fill a recognized gap in the literature.

It would be very good if this initiative were imitated, encouraging other modern protoscientific books to also record and develop their hypotheses in a systematic, collaborative, and revisable way, strengthening a more creative, transparent scientific ecosystem open to future empirical validation.

A.19 Loss of scientific and technological knowledge throughout history

Throughout history, countless pieces of scientific and technological knowledge have been lost, hidden, or delayed for reasons extrinsic to the advancement of knowledge itself. These losses do not stem from human limitation in understanding the world, but from social, economic, political, and cultural factors that interrupt the transmission of knowledge. The main historical causes of this phenomenon are presented below.

A.19.1 Military secrecy

Several advances arose in military contexts and remained secret for decades, whether for war strategy or for reasons of national security. Examples include the development of radar during the Second World War and nuclear and cryptographic projects, such as the *Manhattan Project* and the breaking of the Enigma code. Much of this knowledge was disclosed to the public only after long periods of confidentiality, which delayed its civil and scientific application.

A.19.2 Industrial secrecy

Companies frequently hide innovations to preserve competitive advantages. Proprietary industrial formulas and processes — such as the chemical composition of Coca-Cola — are kept in absolute secrecy. In addition, confidentiality contracts (*Non-Disclosure Agreements*) and reverse-engineering practices hinder the circulation of knowledge and impede the collaborative advancement of certain technologies.

A.19.3 Neglect by the author or by society

Many advances were lost simply through the neglect of the authors themselves or through society's lack of interest. Manuscripts by Archimedes, Hypatia, and other ancient thinkers were destroyed or forgotten. A classic example is that of Gregor Mendel's laws of heredity, rediscovered only thirty-five years after their publication, because they had been ignored in their time.

A.19.4 Decisions by investors or governments

Technologically promising projects are frequently interrupted when there is no expectation of financial or political return. Examples include the abandonment of clean-energy technologies or alternative transport, such as the *pneumatic subway* (Beach Pneumatic Transit) built by Alfred Ely Beach in 1870, which was shelved owing to economic interests and investor pressure.

A.19.5 Patents and legal blocks

The patent system protects the inventor, but it can also inhibit the continuity of scientific development. By restricting the access to and improvement of ideas, patents can paralyze entire areas of research. Large corporations even acquire patents merely to prevent them from being used — a practice known as *patent hoarding*.

An emblematic example of how the patent system can delay technological advancement is that of 3D printers. The principle of additive manufacturing — successive deposition of material to create three-dimensional objects — was developed back in the 1980s, when Charles W. Hull obtained, in 1986, the patent for *stereolithography* (SLA), filed in 1984 (HULL, 1986). For decades, the main 3D-printing techniques remained under the control of a small number of companies, which prevented open development and public access to the technology.

Only after the expiration of the main patents, between 2009 and 2014, did the popularization of 3D printing occur, with the emergence of open-source projects such as *RepRap*. From then on, affordable printers and an explosion of innovations in biotechnology, engineering, and design arose.

This case illustrates how excessive legal protection — even when legitimate — can turn a promising invention into “frozen” knowledge, preventing its collective maturation and its social impact for decades.

A.19.6 Loss of technical or contextual support

Some technologies depend on specific materials, techniques, or historical contexts that could not be reproduced later. The technology of Roman concrete (*opus caementicium*), capable of resisting millennia, remained unknown for centuries. Another example is the Antikythera mechanism, whose mechanical complexity still today challenges the complete understanding of historians of science.

A.19.7 Civilizational collapses and censorship

The physical destruction of centers of knowledge was responsible for incalculable losses. The burning of the Library of Alexandria and the destruction of Mayan and Aztec codices by the European conquerors eliminated centuries of accumulated knowledge. Religious and political censorship also erased entire scientific treatises, interrupting the flow of ideas and the development of theories.

A.19.8 Synthesis

These factors demonstrate that the advancement of science depends not only on human genius, but also on preservation, communication, and freedom of inquiry. The disappearance of a piece of technical or scientific knowledge represents not merely a historical loss, but a collective regression of humanity.

A.19.9 The risk of losing protoscientific knowledge

In the context of this book, *The Metaphor of the Bubbleland*, there exists a real possibility that its publication might never have occurred, owing to the essentially protoscientific character of much of its content. Protoscience, by nature, lies at the frontier between well-grounded speculation and verifiable scientific knowledge; it therefore tends to be rejected by academic institutions and by the formal mechanisms of scientific validation.

However, this initial rejection can have profound consequences. If the propositions formulated here come to be confirmed in the future, the absence of a public record would mean the definitive loss of potentially transformative knowledge. In this sense, the refusal to disseminate protoscientific ideas can function as a form of *epistemic censorship*, preventing certain hypotheses from surviving long enough to be tested.

Thus, the preservation and publication of protoscientific works are justified not merely by their speculative value, but as an act of historical responsibility: ensuring that ideas — however improbable — can remain accessible, so that future generations may revisit them in the light of new instruments, theories, and scientific paradigms.

A.20 Scientific writing, popular writing, and science fiction

Table 3 presents the structural differences between the writing modes of *Scientific publication*, *Science communication*, and *Science fiction (hard)*, under different aspects, highlighting how each form of discourse organizes, expresses, and validates knowledge.

Table 3 – Structural differences between the modes of scientific writing, of science communication, and of science fiction (hard).

Aspect	Scientific publication	Science communication	Science fiction (hard)
Objective	Verify, measure, reproduce	Explain, connect, enchant	Embody human consequences
Evidence	Data, statistics, limits	Analogies + essential facts	Verisimilitude + internal consistency
Language	Technical, impersonal	Metaphorical, accessible	Narrative, symbolic, disciplined
Structure	IMRyD ¹⁴ , references, materials and methods	Metaphors, examples, questions and answers	Scene, character, dramatic arc
Risk	Jargon, opacity	Excessive simplification	Excess of poetic license
Success criterion	Reproducibility, citation	Comprehension without distortion	Emotion without betraying plausibility

A.20.1 Practical guidelines

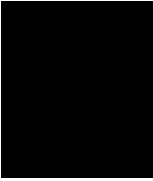
1. **Separate purpose and audience:** define whether the text should prove, explain, or dramatize.
2. **Choose the quality metric:** reproducibility (scientific), intelligibility (communication), dramatic verisimilitude (fiction).
3. **Control the metaphor:** use metaphor to illustrate, not to replace the concept.
4. **Maintain factual anchoring:** even in fiction, establish scientific rules and *follow them* throughout the narrative.
5. **Avoid the “valley of distortion”:** do not simplify beyond the inversion point, at which the metaphor begins to contradict science.

¹⁴ The acronym IMRyD (Introduction, Methods, Results, and Discussion) designates the international standard format for the structure of scientific articles. This model organizes the text into four main sections: *Introduction* (context and hypothesis), *Methods* (experimental description and reproducibility), *Results* (objective presentation of the data), and *Discussion* (interpretation and implications). In English, the most common acronym is IMRAD (*Introduction, Methods, Results and Discussion*).



APPENDIX

B



Types of hypotheses

***The scientist** kills the snake and shows the stick¹.*

***The denialist** says that neither the snake nor the stick exists.*

***The pseudoscientist** says it exists, but shows nothing.*

***The protoscientist** shows the stick, but is still looking for the snake.*

— ANTONIO FERRÃO NETO

THIS is, essentially, a book of hypotheses. Some chapters retain a strong connection with the scientific *status quo*. Others are avowedly metaphorical. There are still others that come close to science fiction, fantasy, or magical realism. But all of them

¹ In the popular Brazilian saying, “to kill the snake and show the stick” (*matar a cobra e mostrar o pau*) means to accomplish a feat and present concrete proof of it.

share something in common: they are thoughts that needed to be laid out — even when they may seem absurd at first sight.

For Science, proposing hypotheses is only the first step: they must be tested, both mathematically and through concrete experiments. The results must be published in peer-reviewed scientific journals. Writing books, giving lectures, offering expert opinions, or posting videos on the internet do not, by themselves, constitute valid sources of scientific research.

If this book dealt only with already-verified hypotheses, they would cease to be hypotheses: they would be theses — and the book would be scientific, not protoscientific.

B.1 Classification of the types of hypotheses

Table 4 presents a classification of different types of hypotheses. This table is not exhaustive, but it is sufficient for the discussions in the text.

Table 4 – Classification of the types of hypotheses

Type	Definition	Example
Promising	Has explanatory potential; deserves investigation.	Certain types of RNA regulate gene expression more than was once thought.
Unpromising	Discredited by evidence and expert opinion.	Diseases are caused by “miasmas” — abandoned after germ theory.
Working	Provisional, used as a basis for testing.	Protein X takes part in metabolic pathway Y.
Denialist	Goes against solid evidence.	Vaccines cause autism.
Absurd	Extremely improbable or illogical.	The universe was created by an invisible duck in 5 minutes.
Falsified	Already refuted, but useful for delimiting scientific progress.	Phlogiston is the principle of fire.
Inconceivable	Cannot be logically formulated, or contradicts reason.	There exists a square circle with three sides.
Strong	Very well grounded and tested.	The Earth revolves around the Sun.
Weak (see Section B.2)	Poorly supported by evidence, but compatible with established facts and potentially testable.	There is microbial life beneath the surface of Mars (as of 2026).

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Type	Definition	Example
Popular	Widely accepted by the majority.	Intelligence has a hereditary component.
Unpopular	Rejected, but may hold hidden value.	Microorganisms cause disease (before Pasteur).
Simple	Easy to understand and test; preferred by Occam's razor.	Plants grow faster with light.
Complex	Involves multiple variables.	Climate change results from the interaction among gases, oceans, and solar activity.
Specific	Applies to a well-delimited phenomenon.	Enzyme Z acts only in the liver of mice.
Generic	Applicable to a variety of situations.	Every effect has a cause.
<i>Ad hoc</i>	Created to save a refuted theory, without independent support.	The planets follow epicycles because that is simply how the universe works.
Exploratory	Seeks to raise ideas, without commitment to immediate confirmation.	Perhaps corals communicate through complex chemical signals.
Predictive	Generates predictions testable in the future.	If string theory is correct, there will be supersymmetric particles.
Metaphysical	Not empirically testable, but relevant to conceptual structure.	There exists a reality beyond spacetime.
Heuristic	Serves as a practical guide for new ideas.	What if the mind were like a computer?
Fanciful	Extremely speculative, with no empirical basis, but internally coherent.	There is a complete universe inside every black hole.
Conjoined	A set of interdependent hypotheses that support a larger one.	The theory of Evolution is supported by genetics, paleontology, and comparative anatomy.
Null	Assumes there is no significant effect or difference.	Drug X does not alter blood pressure.
Alternative	Stands in opposition to the null hypothesis.	Drug X lowers blood pressure.
<i>Ansatz</i>	Initial assumption about the form of a mathematical solution.	We assume that $\psi(x) = Ae^{ikx}$.

B.1.1 Types of inconceivability

The *inconceivable* category can be subdivided into four types:

1. **Logical (absolute):** Formal contradiction. Example: square circle.
2. **Cognitive (neurological):** Permanent mental limitations of certain individuals.
3. **Epistemological (provisional):** Inconceivable at the moment, but comprehensible in principle. Example: quantum effects.
4. **Sociocultural (contextual):** Inconceivable under the dominant paradigms. Example: heliocentrism in the pre-Copernican period.

B.1.2 The taxonomy of hypotheses and methodological pluralism

Table 4 might give the impression that this work adopts a rigid methodological stance, in apparent tension with the pluralism defended by Paul Feyerabend and discussed in Appendix A, Subsection A.4.5. If “anything goes” is the motto of methodological anarchism, how can this openness be reconciled with an apparently restrictive taxonomic classification?

The apparent contradiction dissolves once one recognizes that the two positions operate at *distinct levels* of scientific activity, articulated by the classical distinction between the **context of discovery** and the **context of justification**, formulated by Hans Reichenbach in 1938 (REICHENBACH, 1938).

Context of discovery.

This refers to the process by which hypotheses are *generated*: intuition, analogy, dream, metaphor, productive error, methodological transgression, chance. In this context, Feyerabend is right: there are not, nor should there be, prescriptive rules. “Anything goes” protects scientific creativity against bureaucratic strangulation and recognizes that important discoveries have often arisen from methodological ruptures — Kepler resorting to Pythagorean mysticism, Kekulé dreaming of the structure of benzene, Einstein setting out from thought experiments with no immediate experimental support.

Context of justification.

This refers to the process by which hypotheses, once formulated, are *evaluated*: their internal consistency is examined, their compatibility with evidence is tested,

their epistemic status is classified, their relationship to other hypotheses is mapped. In this context, some form of methodological discipline is indispensable, on pain of losing the capacity to distinguish science from gratuitous speculation.

Where each level operates.

Table 4 operates exclusively in the context of justification: it does not prescribe how hypotheses should be generated, but offers a vocabulary for classifying them after they have arisen, whatever the path of their generation may have been. Feyerabend's pluralism, in turn, operates in the context of discovery: it protects the moment of invention against excessive normativity.

The two levels are therefore complementary. A rigorous protoscientific work needs, simultaneously:

- **Freedom in the context of discovery** — so that metaphors such as that of Bubbleland, weak hypotheses about tunneling into four dimensions, or ontological reinterpretations of entanglement can even be formulated.
- **Discipline in the context of justification** — so that this freedom is not confused with an absence of criteria, so that not every speculation is treated as equivalent, and so that protoscience does not degenerate into rhetorical pseudoscience.

Consequence for reading this work.

Throughout the book the reader will find passages in which the tone is deliberately free — metaphors, thought experiments, analogies with object-oriented programming, speculations about editorial grooves in reality. These passages belong to the context of discovery and should be read with the openness that Feyerabend recommends. In parallel, the reader will find tables, classifications, boxes on scope and limits, explicit declarations of epistemic status. These structures belong to the context of justification and fulfill the disciplinary function that the Popperian, Kuhnian, and Lakatosian tradition justifies.

There is no contradiction between the two modes: there is a functional division of labor. The book can be at once bold in its propositions and rigorous in its classification, because boldness and rigor operate in different layers of the same process. To confuse them would be a category error; to separate them is a condition for protoscience to function as a legitimate stage, and not as a disguise.

Synthesis.

Methodological pluralism describes how science *is born*; the taxonomy of hypotheses describes how science *is organized once it has been born*. This work adopts both, each at its own level. Feyerabend and Popper, Kuhn and Lakatos, need not be chosen as mutually exclusive: they need to be correctly allocated to the distinct phases of the production and the evaluation of knowledge.

Connection with the black swan taxonomy.

The articulation between the context of discovery and the context of justification operates, at a higher plane, in dialogue with the meta-methodology developed in Appendix AB (Vol. III). There, a taxonomy of four stances toward scientific anomalies is proposed — from *structured denial* (Type I), through *silent perplexity* (Type II) and *declared astonishment* (Type III), up to *methodological openness* (Type IV) — and the stance adopted by this work is explicitly identified as belonging to Type IV.

The relationship between the two taxonomies is one of functional complementarity. The taxonomy of hypotheses of the present appendix classifies *what* is proposed: it distinguishes legitimate weak hypotheses, conjoined ones, *ad hoc* ones, falsified ones, among other categories. The black swan taxonomy of Appendix AB (Vol. III) classifies *how* one reacts to that which resists theoretical domestication: it distinguishes cowardly, perplexed, honest, and methodologically open stances. A legitimate weak hypothesis, in the sense of the criteria Cr1–Cr5 of Section B.2, is typically the product of a Type IV stance toward an anomaly: the hypothesis exists because the black swan was welcomed without being painted white.

Together, the two taxonomies form a two-dimensional apparatus: the *content* axis (what type of hypothesis?) and the *attitude* axis (what stance toward the anomaly that gave rise to it?). Only the conjunction of the two axes makes it possible to distinguish, with precision, a legitimate weak hypothesis from a disguised *ad hoc* hypothesis — a distinction essential to prevent protoscience from degenerating into rhetorical pseudoscience.

B.2 Admissibility criteria for weak hypotheses

Table 4 includes, among the possible categories, the **weak hypothesis** — that is, one that is poorly supported or insufficiently defined. This classification is central to the

present work, since many of the propositions discussed throughout the chapters are explicitly presented as weak hypotheses. A methodological question therefore arises that cannot be avoided: *what distinguishes a legitimate weak hypothesis from a pseudoscientific claim disguised under the same label?*

Without a clear answer, the expression “weak hypothesis” runs the risk of becoming a rhetorical shield — a way of proposing anything without epistemic commitment, invoking its own fragility as a justification for the absence of evidence. This practice would bring protoscience closer to pseudoscience, subverting precisely the boundary the work seeks to preserve.

7. Definition of protoscience

Protoscience is a field of inquiry at an embryonic — pre-paradigmatic, in Kuhn’s sense (KUHN, 1962) — stage, which adopts the principles of the scientific method and seeks to test its hypotheses empirically, having the potential to consolidate into established science, but without any prior guarantee of that outcome. It is in this sense that Appendix A situates this work (Section A.6), it is to this sense that the criteria Cr1–Cr5 below give operational form, and it is this sense that the entry “protoscience” in the Glossary summarizes (Volume I, Page 367).

Five minimum admissibility criteria are proposed below that a weak hypothesis must meet in order to be considered legitimate in the protoscientific sense adopted in this book. The criteria are designated **Cr1** through **Cr5** (short for *admissibility criterion*), distinct from the abbreviation **C** reserved for *corollaries* in other chapters.

B.2.1 Five admissibility criteria

- Cr1. Internal consistency.** The hypothesis cannot contain explicit logical contradictions, nor require, in order to be maintained, the simultaneous negation of premises that it itself assumes. This is the most basic criterion and corresponds to the *logical inconceivable* type of Section B.1: a hypothesis that is born self-contradictory is not weak — it is inadmissible.
- Cr2. Compatibility with established facts.** The hypothesis cannot require, as a precondition, the falsification of experimental results already robustly corroborated. A weak hypothesis may *extend*, *reinterpret*, or *complement* established knowledge, but it cannot contradict it without offering an explicit mechanism that explains why the earlier evidence should be revised. This criterion

distinguishes legitimate weak hypotheses from *denialism*.

- Cr3. Generativity.** The hypothesis must produce consequences — predictions, implications, constraints on observables, points of contact with other theories — even if those consequences are not immediately testable with the available technology. A weak hypothesis that generates no differential consequence is not protoscientific: it is pure metaphysics. This criterion incorporates the Popperian requirement of empirical content, but in an attenuated form: immediate testability is not demanded, only the *potential* for future testability.
- Cr4. Transparency of status.** The author of a weak hypothesis must explicitly declare its epistemic status and must not present it as stronger than it is. This criterion is as important as the others, since it distinguishes protoscientific honesty from the rhetorical simulation practiced by pseudoscience. Table 14 (Vol. I) of Chapter 18 (p. 353) performs precisely this function for the set of hypotheses in this work.
- Cr5. Refutability in principle.** The hypothesis must be, at least in principle, vulnerable to contrary evidence — under some imaginable technology, instrument, or theoretical framework. A weak hypothesis that is structurally immune to any conceivable refutation leaves the field of protoscience and enters that of untestable metaphysics. This criterion admits gradation: hypotheses whose refutation would require technology that does not exist today are acceptable, provided the refutation remains logically possible.

B.2.2 Application: legitimate weak hypothesis *versus* pseudoscientific hypothesis

Table 5 illustrates how the five criteria separate legitimate weak hypotheses from claims that merely present themselves as such.

Table 5 – Application of the admissibility criteria to exemplary cases.

Example	Cr1	Cr2	Cr3	Cr4	Cr5	Classification
Tunneling of biomolecules into a fourth spatial dimension (Ch. 8, Vol. I)	✓	✓	✓	✓	✓	Legitimate weak hypothesis
Inter-reality class variable for entanglement (Subsection 17.4.7, Vol. I)	✓	✓	partial	✓	partial	Deliberately weak generative metaphor
Vaccines cause autism	✓	×	✓	×	✓	Denialism: falsified and maintained (not a weak hypothesis)
Some people live without eating or drinking (inedia)	×	×	partial	×	✓	Pseudoscience
There is microbial life on Mars (before 2026)	✓	✓	✓	✓	✓	Legitimate weak hypothesis (still open)
Globally rotating universe (SZIGETI et al., 2025)	✓	partial	✓	✓	✓	Weak hypothesis at the limit of admissibility

Three observations follow from the table:

1. A hypothesis may partly meet a criterion without thereby ceasing to be admissible as weak, provided the remaining criteria compensate. Epistemic strength is the joint result, not the sum of individual criteria.
2. Criterion Cr4 (transparency) is the only one in which failure is *qualitative* rather than *quantitative*: a hypothesis presented with the wrong status ceases to be protoscience, regardless of whether it meets the other criteria.
3. The classification of a hypothesis may change over time. The hypothesis of microbial life on Mars was weak in 2000; it remains weak in 2026, but closer to testability. The hypothesis of continental drift was weak from 1912 to 1960 and today is established theory. The *weak* status is historical, not ontological.

B.2.3 Methodological consequence for this work

Each hypothesis classified as “weak” (**W**) in Table 34 (Vol. III) of Appendix AH (p. 391) was subjected to the five criteria **Cr1–Cr5** above before being admitted into the body of the book. Making these criteria explicit in the present appendix does not

introduce a new filter: it merely formalizes, in methodological language, the scrutiny that has guided the composition of the work from the start.

This filter is, in the last analysis, what justifies the editorial existence of a proto-scientific work. Without it, any speculative claim could lay claim to the label “weak hypothesis” and, with it, also claim a legitimacy it does not possess. With it, proto-science keeps itself in a well-defined intermediate position: freer than established science, more disciplined than free speculation.

B.2.4 Sweep of the weak hypotheses of the work

Table 6 presents the individual application of the criteria **Cr1** through **Cr5** to each of the 34 hypotheses classified as weak (**W**) in Table 34 (Vol. III) of Appendix AH (p. 391). The purpose is twofold: (i) to verify that no approved hypothesis silently violates any of the criteria; (ii) to offer the reader an auditable record of the classificatory process.

Marking convention.

✓ indicates full compliance; *partial* indicates compliance with substantive reservations discussed in what follows; ✓[†] indicates compliance only in principle, with no foreseeable experimental path within the current technological horizon; × would indicate failure and consequent reclassification.

Table 6 – Sweep of the admissibility criteria Cr1–Cr5 applied to the weak hypotheses of the work.

ID	Hypothesis (summary)	Cr1	Cr2	Cr3	Cr4	Cr5
H01	Realities coexist in parallel planes separated by additional dimensions	✓	✓	✓	✓	✓
H03	5D manifold foliated by parallel 4D realities	✓	✓	✓	✓	✓
H04	Parallel realities may be non-geometric (informational, consciousness-based)	✓	✓	partial	✓	✓ [†]
H07	Nonlinear electric field in the high-intensity regime near the electron	✓	✓	✓	✓	✓
H08	Strong and weak interactions as effective nonlinear regimes	✓	✓	✓	✓	✓
H09	Quantization of light via nonlinear saturation of the field (alternative to Planck)	✓	partial	✓	✓	✓

Continued on next page...

Table 6 (continued)

ID	Hypothesis (summary)	Cr1	Cr2	Cr3	Cr4	Cr5
H12	Electromagnetic and gravitational subnomies are ontologically primitive	✓	✓	✓	✓	✓
H13	The 1s orbital as a distribution of epistemic ignorance, not real motion	✓	✓	partial	✓	partial
H14	Biomolecules access 4D degrees of freedom under specific conditions	✓	✓	✓	✓	✓
H15	Biomolecules transition to a parallel reality of complementary nomy	✓	✓	✓	✓	✓
H16	DNA accesses 4D to resolve topological knots	✓	✓	✓	✓	✓
H17	Microtubules take part in the physical substrate of consciousness	✓	✓	✓	✓	✓
H18	Continuous coupling between our reality and the complementary one at the biomolecular scale	✓	✓	✓	✓	✓
H19	Time in parallel realities detuned from our own	✓	✓	✓	✓	✓
H20	Radioactivity as a probe for temporal detuning	✓	✓	✓	✓	✓
H21	Time travel creates new lines (without the grandfather paradox)	✓	✓	partial	✓	✓ [†]
H22	Non-intrusive probes monitor the time tree passively	✓	✓	partial	✓	✓ [†]
H23	The human mind operates partly in a domain distinct from the biological brain	✓	✓	✓	✓	✓ [†]
H24	Box jellyfish perform visual processing external to their nerve ganglia	✓	✓	✓	✓	✓
H29	Quantum computing accelerates the discovery of nomies (Quantum SINDy)	✓	✓	✓	✓	✓ [†]
H32	Self-replicating SZM devices at the molecular scale viable in principle	✓	✓	✓	✓	✓ [†]
H33	Quantum entanglement reflects a class variable in a parallel reality	✓	✓	partial	✓	partial
H34	Transversal causality allows interventions without a path in the observable Plane	✓	partial	partial	✓	✓ [†]
H35	“Hyperevolution”: abrupt evolutionary jumps via inter-reality coupling	✓	✓	✓	✓	✓
H36	Dark matter belongs to subnomies or inaccessible parallel realities	✓	✓	✓	✓	✓
H37	Dark energy has a nomological origin in a parallel reality	✓	✓	✓	✓	✓ [†]
H38	Primordial biomolecules tunneled, resolving the RNA paradox	✓	✓	✓	✓	✓ [†]
H39	Parallel realities define the behavior of our biomolecules	✓	✓	✓	✓	✓
H40	Fundamental Lagrangian merely as an algorithmic procedure	✓	✓	✓	✓	✓ [†]

Continued on next page...

Table 6 (continued)

ID	Hypothesis (summary)	Cr1	Cr2	Cr3	Cr4	Cr5
H41	19 free parameters of the SM determined by the geometry of the solitons	✓	✓	✓	✓	✓ [†]
H42	3 generations of fermions as radial excitations or topological sectors	✓	✓	✓	✓	✓ [†]
H43	Interior of the solitons: full gauge symmetry restored	✓	✓	✓	✓	✓ [†]
H44	Pauli exclusion emerges from “onion”-type geometric nonlinearities	✓	✓	✓	✓	✓ [†]
H45	Parallel 3D realities coexist along a fourth spatial dimension v	✓	✓	✓	✓	✓

Overall verdict.

None of the 34 weak hypotheses failed any criterion (there is no $\times$ in the table). The “partial” and “[†]” markings signal that some hypotheses meet the criteria with reservations that, although they do not compromise their admissibility, deserve to be made explicit. The non-trivial cases are discussed below.

H04 (non-geometric realities).

The *Testability* column of Table 34 (Vol. III) explicitly records “conceptual test; no direct experimental protocol”. The hypothesis meets **Cr5** only in principle: it is refutable by a logical demonstration of the incoherence of the very concept of non-geometric reality, but not by any foreseeable empirical experiment. The “partial” marking in **Cr3** reflects that its generativity is predominantly conceptual — the hypothesis stimulates philosophical distinctions, but generates few consequences with immediate empirical contact.

H09 (quantization via nonlinear saturation).

The reservation on **Cr2** arises from the fact that the hypothesis proposes an alternative explanation for an extremely well-established empirical result (the photoelectric effect, blackbody spectra, photon counting). It is only compatible with the facts *if* it quantitatively reproduces, in the appropriate regime, the predictions of Planck’s quantization. Compatibility is therefore conditional: Appendix M assumes

this reproduction as a burden to be discharged, but the explicit derivation has not yet been carried out.

H13 (the 1s orbital as epistemic ignorance).

It meets **Cr1–Cr2–Cr4** without reservations, but its generativity (**Cr3**) is partial: by construction, the reinterpretation does not alter the predictions of quantum mechanics, which is simultaneously its virtue (total compatibility) and its limitation (absence of distinctive empirical consequences). The same applies to **Cr5**: the hypothesis is refutable only indirectly, through the refutation of the broader nomological framework in which it is embedded. It is the archetypal case of *ontological reinterpretation without independent observational consequence* — legitimate, but epistemically modest.

H21 and H22 (temporal branching and passive probes).

Both meet **Cr5** only in principle. Branching without the grandfather paradox is refutable by a demonstration of internal logical inconsistency (not yet found) or by the empirical observation of genuine paradoxes (inaccessible). The non-intrusive probes depend on nonexistent technology with no foreseeable horizon. The “partial” marking in **Cr3** for both reflects that their generativity is structural (compatibilization with Many-Worlds) rather than predictive.

H23 (mind in a domain distinct from the brain).

It meets **Cr5** only in principle. The differential consequence *does exist* and is stated in Subsection 5.7.4.3 (Vol. I): an experiment that would produce different results depending on which hypothesis is true — detection of residual conscious activity in brains with damage that, under the production hypothesis, should eliminate it completely; or a demonstration that certain aspects of consciousness are insensitive to global brain damage. The chapter itself qualifies both as “experimentally difficult, but not incoherent”, and it is that distance between logical refutability and an experimental path that the † marking records. The objection cataloged in Appendix J — that the hypothesis would generate no falsifiable predictions — finds its answer here, and there.

H29, H32, H37, H38, H40–H44 († on Cr5).

This set shares a common characteristic: refutability in principle is fully assured, but depends on technological advances (large-scale quantum computing, self-replicating SZMD, a viable SZM, observational access to the origin of life) whose horizon is distant or indeterminate. The † marking does not weaken the admissibility of these hypotheses; it merely makes it auditable that their test is deferred to a future that is not guaranteed. In particular, **H40–H44** form Cluster F of Appendix AH (Vol. III), whose testability is *critically dependent* on the viability of the SZM (H30); this dependence is already made explicit in the hypothesis mapping itself.

H33 (inter-reality class variable) and H34 (transversal causality).

These are the two hypotheses with the greatest number of reservations, and they deserve special attention. H33 fully meets **Cr1–Cr2–Cr4**, but its generativity and refutability are partial: the hypothesis offers an *alternative language* compatible with the Bell correlations, without experimental predictions distinctive from the standard formulation. H34 is the most delicate case: its compatibility with established facts (**Cr2**) is “partial” because the notion of causality transversal to the observable timeline strains (without formally contradicting) the principle of local causality; its generativity (**Cr3**) is restricted to providing language for rare phenomena categorized as “editorial grooves”; and **Cr5** is met only in principle. Table 34 (Vol. III) itself describes H34 as a “metaphysical hypothesis; no experimental protocol”. Its retention in the body of the book is justified by its structural function in Cluster E and by its declared status as a generative metaphor, not an empirical conjecture. This nature of H33 is the same one already recorded in Table 5 under the label “deliberately weak generative metaphor” — the two tables are therefore consistent with each other: Table 5 gives the qualitative label, and Table 6 gives the granular sweep of the criteria.

Conclusion of the sweep.

No reclassification is necessary. The work retains the 34 weak hypotheses originally classified as such, with the following explicit reservation markings to be incorporated into future editions of Table 34 (Vol. III) — possibly as an additional

column “Cr5 technologically accessible” (Yes/No/In principle): H04, H21, H22, H23, H29, H32, H34, H37, H38, H40, H41, H42, H43, H44. These fourteen hypotheses remain protoscientific, but their test horizon is distant; the reader should therefore interpret them as *research horizons* rather than as *conjectures in immediate empirical dispute*. The remaining 17 weak hypotheses (34 in total minus the 17 carrying some reservation marking — † or “partial”) meet the five criteria without reservations and constitute the set of *weak hypotheses effectively at empirical risk* in the strong Popperian sense.

B.3 Mutual reinforcement of related weak hypotheses

A single arrow breaks easily, but a bundle of ten arrows is invincible.

— JAPANESE PROVERB

Section B.2 established the criteria **Cr1–Cr5** applicable to weak hypotheses considered *individually*. There remains the complementary question to be examined: *under what conditions can a set of individually weak hypotheses acquire, collectively, epistemic strength greater than the sum of its parts?*

A set of individually weak hypotheses may, in certain circumstances, acquire greater epistemic strength when these hypotheses are articulated into a coherent network. This does not mean that the mere accumulation of conjectures produces theoretical solidity, but that the relationship among them can generate an effect of mutual reinforcement when there is explanatory convergence, logical compatibility, and heuristic fecundity.

B.3.1 Central idea

A weak hypothesis, taken in isolation, may seem insufficient because it presents little direct evidence, low predictive power, or scant connection with other known facts. However, when several hypotheses of this kind:

- address different aspects of the same problem;
- are logically compatible with one another;
- explain partially independent phenomena;
- point toward the same underlying structure;

a theoretical architecture may emerge that is more robust than the isolated assessment of each one would suggest.

The strength of the set, in this case, does not follow merely from the sum of the parts, but from the fact that each hypothesis comes to occupy a functional role in a broader explanatory network. In epistemological terms, plausibility may migrate from the isolated element to the structure of relations that connects it to the others (BONJOUR, 1985).

B.3.2 Consilience of evidence

One of the classic mechanisms of this reinforcement is consilience, a concept introduced by Whewell (WHEWELL, 1840). It occurs when different lines of evidence, none of them decisive on its own, converge toward the same theoretical conclusion.

In this case, each hypothesis or sub-hypothesis illuminates a distinct region of the problem. One may explain an empirical pattern, another may clarify a conceptual constraint, another may supply a possible mechanism, and another may connect the system to already-established theories. In isolation, each remains fragile; together, they reduce the impression of arbitrariness.

Consilience is especially important when:

- the data are incomplete;
- no crucial experiment is available;
- the object under study is partially inaccessible;
- the theory is still in an exploratory phase.

In these situations, the convergence of independent lines of evidence may be more significant than the brute force of a single piece of evidence.

B.3.3 Structural coherence and theoretical networks

Hypotheses do not, in general, live alone. In scientific practice, they usually appear embedded in theoretical networks made up of definitions, auxiliary principles, analogies, idealizations, proposed mechanisms, and mathematical constraints. This perspective is compatible with Lakatos's notion of *research programmes* (LAKATOS, 1978), according to which individual hypotheses are evaluated in the context of broader theoretical structures, endowed with a hard core and a protective belt of auxiliary hypotheses.

A weak hypothesis may gain importance when it:

- fills a conceptual gap;
- connects two previously disconnected parts of the theory;
- eliminates internal tensions in the system;

- makes it possible to derive new consequences;
- renders intelligible a previously scattered set of phenomena.

In this sense, rational evaluation bears not only on the immediate truth of the hypothesis, but also on its architectural function. A proposition that is not yet well confirmed may deserve attention if its removal would destroy the intelligibility of the whole picture.

B.3.4 Inference to the best explanation

Another important point is the role of inference to the best explanation, a notion systematized by Harman (HARMAN, 1965) and developed in depth by Lipton (LIPTON, 2004). Suppose there are several scattered facts, each poorly elucidated by independent explanations. If an articulated set of hypotheses manages to explain them simultaneously, then the system as a whole gains plausibility.

This gain in plausibility may derive from criteria such as:

- unification;
- relative structural simplicity;
- explanatory power;
- economy of assumptions;
- theoretical fertility;
- the capacity to suggest future tests.

These criteria correspond, to a large extent, to the *theoretical virtues* discussed by Kuhn (1977) and to the criteria of theory choice analyzed by Thagard (THAGARD, 1978). The decisive point is that strength does not necessarily lie in an isolated proof, but in the comparative superiority of the theoretical picture over rival alternatives. Thus, several related weak hypotheses may, together, form the best available explanation at a given stage of the investigation.

B.3.5 Probabilistic analogy: Bayesian intuition

This phenomenon can also be understood through a probabilistic analogy inspired by Bayesian reasoning (HOWSON; URBACH, 2006; JAYNES, 2003). The central idea is that the confirmation of a hypothesis depends not only on its prior plausibility, but also on the way in which the evidence rationally alters that plausibility.

Let $H_1, H_2, \dots, H_n$ be related hypotheses, and let E be a set of evidence. In general

terms, Bayesian reasoning suggests that:

$$P(H \mid E) = \frac{P(E \mid H) P(H)}{P(E)}.$$

That is, the posterior probability of a hypothesis H , given the evidence E , depends on:

- the prior plausibility $P(H)$;
- the expectation of observing E if H is true, that is, $P(E \mid H)$;
- the overall probability of the evidence $P(E)$.

When the hypotheses are related, the situation becomes more interesting. The precise mechanism by which the evidence propagates among them lies in the **structure of the joint prior distribution**. If the hypotheses are correlated a priori — that is, if $P(H_2 \mid H_1) > P(H_2)$ — then evidence favorable to H_1 raises the posterior probability of H_1 and, through this prior correlation, indirectly raises the plausibility of H_2 . In more direct terms: mutual reinforcement does not arise only at the stage of updating on evidence, but is *inscribed* in the network of dependencies that connects the hypotheses even before any observation. In modern language, this structure can be represented by **Bayesian networks**, in which the nodes correspond to hypotheses and the edges represent probabilistic dependencies (PEARL, 2009; DARWICHE, 2009).

For example, if:

- H_1 proposes the existence of a certain hidden structure;
- H_2 proposes a mechanism compatible with that structure;
- H_3 proposes an observable effect derived from that mechanism;

then evidence favorable to H_3 may rationally propagate back onto H_2 and H_1 , provided the dependence among them is genuine and not artificial.

In schematic terms, instead of evaluating only isolated probabilities such as $P(H_1)$, $P(H_2)$, and $P(H_3)$, one comes to consider joint distributions, such as:

$$P(H_1, H_2, H_3 \mid E).$$

If the hypotheses form an integrated structure, it may happen that:

$$P(H_1, H_2, H_3 \mid E)$$

becomes epistemically more significant than the separate analysis of each term. In other words: the system may become more credible as an articulated configuration than as a collection of independent propositions.

However, this intuition works only when there is **genuine rational dependence** among the hypotheses — that is, dependence that emerges from real causal, nomological, or structural relations. If, on the contrary, the connection is merely **stipulated** by the proponent — constructed *ad hoc* for argumentative purposes, without anchoring in independent constraints — the appeal to joint probability does not strengthen the theory; it merely masks its fragility. This distinction between discovered dependence and fabricated dependence is decisive if the Bayesian analogy is not to be reduced to a rhetorical device.

B.3.6 The effect of mutual constraint

An important aspect, sometimes neglected, is that related hypotheses do not merely support one another; they also constrain one another. This can be a virtue. When one hypothesis limits the space of possibilities of the others, the set may become less vague and more controllable.

For example:

- an ontological hypothesis may constrain the kinds of admissible mechanisms;
- a mathematical hypothesis may eliminate conceptually loose interpretations;
- an empirical hypothesis may discard extreme versions of another hypothesis;
- a methodological hypothesis may prevent arbitrary extrapolations.

This effect of internal binding reduces excessive degrees of freedom. A system with less arbitrariness tends to have more epistemic value than a set of conjectures that can accommodate any result.

B.3.7 Heuristic fecundity

Even before being strongly confirmed, related hypotheses may gain relevance if the set is heuristically fertile. This means that the theoretical network:

- suggests new experiments;
- guides the collection of data;
- produces more precise questions;
- makes it possible to distinguish rival scenarios;
- indicates where to look for anomalies or observable signatures.

This fecundity does not prove the theory, but it increases its rational interest. At many moments in the history of science, still-fragile theories were taken seriously because they opened up promising research trails (LAKATOS, 1978).

B.3.8 Limits of mutual reinforcement

Caution is required. Not every set of weak hypotheses forms a strong theory. There are situations in which aggregation produces only an illusion of robustness.

Mutual reinforcement fails when:

- the hypotheses are too vague;
- the connections among them are merely verbal;
- there are internal contradictions;
- the system accommodates any evidence;
- there are no minimally clear criteria of testing, refutation, or comparison;
- the network is constructed *ad hoc* solely to protect a central idea.

In these cases, what looks like reinforcement is merely rhetorical armor. Instead of a rationally integrated structure, there is an accumulation of speculations.

Moreover, it is worth recognizing that the evaluation of theoretical networks does not take place in a social vacuum. Institutional and psychological factors — such as academic incentives, collective confirmation biases, effects of authority, and group dynamics in research communities — can make a network of weak hypotheses *seem* stronger than it really is, simply because a community has invested in it. Contemporary social epistemology, in the wake of works such as those of Kuhn (1962) and Longino (LONGINO, 1990; LONGINO, 2002), warns that the perceived coherence of a theoretical system may reflect, in part, the sociological consensus of its proponents, and not only the objective quality of its internal relations. This dimension does not invalidate the argument for mutual reinforcement, but it demands additional vigilance: the evaluator must ask whether the impression of robustness survives once the effects of group belonging and prior investment are discounted.

B.3.9 Empirical underdetermination

Even when a network of hypotheses satisfies all the criteria above — coherence, consilience, mutual constraint, fecundity — a background difficulty remains: the problem of empirical underdetermination, classically associated with Duhem (DUHEM, 1906) and Quine (1951). The point is that, in principle, there may be multiple theoretical networks that are equally coherent, convergent, and fertile, all compatible with the same available data.

If this happens, the internal coherence of the system — however impressive it may be — is not sufficient to determine its truth. It reduces the space of candidate theories, but does not necessarily reduce it to a single option.

This point does not invalidate the argument developed in the previous sections. Consilience and structural coherence remain genuine epistemic virtues. It does, however, add a layer of humility: the mutual reinforcement of hypotheses increases the relative plausibility of a system, but does not, by itself, guarantee that this system is the only one or the definitive one. The possibility of equally well-articulated rival alternatives must remain a regulative horizon of the investigation.

B.3.10 Criteria of legitimate strengthening

One may say that related weak hypotheses strengthen one another legitimately when the set exhibits, at least to an initial degree, the eight criteria designated **Cf1–Cf8** (*strengthening criteria*), distinct from the criteria of individual admissibility **Cr1–Cr5** established in Section B.2. Each criterion is presented below with an operational definition and an applicable *test-question* for verifying that it is met.

Cf1. Internal logical coherence. *Definition:* the hypotheses of the set do not contradict one another, nor do they simultaneously require the truth and the falsity of one and the same proposition. It differs from **Cr1** (internal consistency of *each* hypothesis) by operating at the level of the *network*: each hypothesis may be internally consistent and yet contradict another in the set.

Test-question: “is there any combination of hypotheses in the cluster which, if all were true, would generate a formal contradiction?”

Cf2. Conceptual compatibility. *Definition:* the ontological, methodological, or semantic assumptions of the hypotheses are mutually accommodable, even when they are not logically derived from one another. It differs from **Cf1** because it concerns *tacit compatibility* (vocabulary, categories, scope), and not explicit contradiction. Two hypotheses may be logically compatible (Cf1 met) and conceptually dissonant (Cf2 violated) — for example, one assuming structural realism and the other radical instrumentalism.

Test-question: “is it possible to hold all the hypotheses of the set without switching ontological or methodological framework between them?”

Cf3. Consilience. *Definition:* independent empirical or theoretical lines of evidence — that is, arising from distinct phenomena, scales, or domains — converge toward the same explanatory picture. The greater the independence of the sources of evidence, the greater the consilient strength. The concept goes back to Whewell (WHEWELL, 1840) and requires that the evidence be not merely

multiple, but *causally disconnected* from one another.

Test-question: “does the evidence supporting different hypotheses of the cluster come from domains sufficiently distinct that their convergence cannot be explained by a trivial common cause?”

Cf4. Superior explanatory power. *Definition:* the set explains phenomena that articulated rival alternatives (not mere negations) explain worse, with more auxiliary hypotheses or with less unification. It requires the *nominal* identification of the alternatives compared; a set cannot claim superiority against an unspecified adversary.

Test-question: “what are the two or three best-articulated rival alternatives, and does the proposed set really surpass them in unification, parsimony, or empirical coverage — or does it merely ignore them?”

Cf5. Non-trivial mutual constraints. *Definition:* accepting one hypothesis of the set *limits* the space of admissible formulations of the others. The hypotheses are not merely compatible (Cf2) or logically coexistent (Cf1): each one *constrains* what the others can assert, reducing the degrees of freedom of the system. It differs from **Cf3** (consilience) because consilience concerns *convergence of evidence*, whereas mutual constraint concerns *reciprocal limitation of formulations*.
Test-question: “if I modified one hypothesis of the cluster, how many others would also have to be modified to preserve coherence? If the answer is ‘none’ or ‘almost none’, the mutual constraints are trivial.”

Cf6. Heuristic fecundity. *Definition:* the set suggests new questions, new experiments, or new connections between domains that *would not be formulated* from the isolated hypotheses, nor from the rival alternatives. The fecundity may be empirical (suggesting tests), theoretical (suggesting unifications), or methodological (suggesting instruments).

Test-question: “does the set generate at least three concrete, investigable questions that were not available before its articulation?”

Cf7. Exposure to empirical or theoretical risk. *Definition:* at least part of the set admits, in principle, conditions under which it would be refuted — experimental, observational, logical, or mathematical. It differs from **Cr5** (individual refutability) by operating at the level of the set: the refutation may bear on the cluster as a whole (for example, failure to derive a key prediction that articulates several hypotheses), and not only on a specific hypothesis.

Test-question: “is there an experiment, observation, or mathematical proof — currently or conceivably in the future — whose realization would render the cluster, as a whole, untenable?”

Cf8. Resistance to analysis with sociological factors discounted. *Definition:* the perceived coherence of the set does not depend essentially on the consensus of a community of researchers that has invested in it. When the effects of group belonging, authority, collective confirmation bias, and prior investment are discounted, the set still appears robust. This is a collective version of the epistemological vigilance discussed in Section B.3.8.

Test-question: “would an external evaluator, with no institutional or affective tie to the proponents of the cluster, reach similar conclusions about its coherence? Or does the perceived robustness depend substantially on who is doing the looking?”

Gradation in compliance.

The criteria admit three levels of compliance, following the convention adopted in Table 35 (Vol. III) of Appendix AH (p. 391): **fully met** (✓), **partially met** (*partial*), and **failed** (×). Failure on any Cf does not automatically invalidate the cluster — unlike Cr, in which failure implies inadmissibility — but the accumulation of failed or partial Cf progressively reduces the legitimacy of the mutual reinforcement, bringing the set closer to the *rhetorical armor* discussed in Subsection B.3.8.

The more Cf are present to a full degree, the greater the chance that the individual fragility of the parts will be compensated by the relational robustness of the set. The presence of **all eight** to a full degree, however, is unlikely outside mature science: legitimate protoscientific programs typically meet Cf1, Cf2, Cf3, and Cf6 with ease, Cf4, Cf5, and Cf7 partially, and hardly ever Cf8 before institutional consolidation.

B.3.11 Articulation between cr and cf

The two systems of criteria operate at distinct and complementary levels:

- **Cr1–Cr5** are *necessary conditions* for each hypothesis to even be admitted into the protoscientific program. A hypothesis that fails any Cr is not a legitimate weak hypothesis — it is inadmissible.

- **Cf1–Cf8** are *conditions of collective potentiation*: they describe how already-admitted hypotheses (those that satisfy Cr1–Cr5) can, when articulated into a network, acquire plausibility greater than the sum of the parts.

The satisfaction of **Cr1–Cr5** is *logically prior* to the application of **Cf1–Cf8**: it makes no sense to evaluate the mutual reinforcement of inadmissible hypotheses. On the other hand, **Cf1–Cf8** do *not replace* **Cr1–Cr5**: a set may satisfy all the Cf and still contain an individual hypothesis that violates some Cr, in which case the problem is localized and requires the reclassification of that specific hypothesis.

B.3.12 Synthesis

Therefore, a set of weak hypotheses can indeed gain strength when it is organized as a coherent, convergent, and fertile network. The emergent solidity does not come magically from the quantity of hypotheses, but from the quality of the relations among them. The central point is not merely to ask whether each hypothesis is strong in isolation, but whether the system they form constitutes an explanation that is more intelligible, more constrained, more unified, and more promising than its alternatives — without losing sight of the fact that coherence, although necessary, may not be sufficient in the face of the underdetermination and the social biases that accompany every theoretical enterprise.

In short: weak hypotheses do not become true simply by being together; but they may, taken together, become rationally more considerable than they would be separately. It is this possibility of relational strengthening that justifies, in many contexts, the effort of building theoretical networks even when their individual components are still at an exploratory stage — provided one maintains, in parallel, the critical awareness that this increased consideration does not, by itself, amount to confirmation.

8. Key idea: where the bundle stops holding

The metaphor of the bundle — arrows, reeds, or pieces of evidence — has a precise limit, and it is worth stating before the reader meets it unaided: it holds for propositions and fails for people.

The reason is not moral but logical. Hypotheses, clues, and proofs can be aggregated without violence because they have no interiority: they exist to be classified, compared, and gathered. A subject, by contrast, has consciousness, will, and a project of their

continued →

8. Key idea: where the bundle stops holding (*continued*)

own, and “binding” them to a set requires suppressing precisely what constitutes them. To apply to the domain of subjects the logic that holds for the domain of objects is a **category error**, and it is from this error that every doctrine treating the individual as a piece of a block is born.^a

Note that the Roman symbol preserves exactly what the doctrine destroys: in the bundle, each rod remains *whole*, and the strength results from the sum of intact units. A collective that holds together only by planing down its parts does not fulfil the metaphor — it negates it.

In the epistemological domain the bundle also has a limit, subtler and already recorded in **Cf3**: mutually reinforcing pieces of evidence may share a single tainted origin, and a bundle of non-independent clues firmly sustains a false conclusion. The difference between the two limits is that the epistemological one is correctable.

^a It is from this image that the best known of those doctrines takes its name. The Roman *fascēs* was a bundle of birch or elm rods, bound with straps around an axe and carried on the shoulder by the lictors who preceded the magistrate, as the insignia of his authority. Within Rome the axe was removed from the bundle: without it, the symbol declared that the citizen could appeal a death sentence (BRENNAN, 2022).

B.4 Stages of a scientific hypothesis

1. Untested hypothesis

Every scientific hypothesis begins as untested. Some remain indefinitely at this stage, whether because of practical difficulties or because they function as *working hypotheses*.

2. Inconclusive hypothesis

Even after several attempts at testing, a hypothesis may remain without a definitive conclusion, owing to instrumental limitations or to the complexity of the phenomenon.

3. Falsified hypothesis

Falsification does not represent definitive exclusion. Previously falsified hypotheses may be reconsidered in the light of new data. Example: the corpuscular hypothesis of light, considered inadequate after Young, was revived by Planck and Einstein.

4. Corroborated hypothesis

The central aim of science is to build hypotheses that withstand rigorous

testing. According to Popper, even the best-supported hypotheses always remain subject to future refutation.

Note: The transition between these stages is not linear. A hypothesis may return to earlier stages as new methods or data emerge.

B.5 When should one believe a hypothesis?

In science, a hypothesis is accepted only when it systematically withstands rigorous attempts at refutation. At that moment, it ceases to be a hypothesis and becomes established knowledge.

Good hypotheses:

- Generate many testable consequences
- Offer explanatory or predictive promise
- Follow Occam's Razor, avoiding unnecessary assumptions

Bad hypotheses: They may seem sterile, but they often help to reveal the limits of current models or to inspire reformulations.

Denialist hypotheses: They reject well-established evidence without stronger grounds. They should be discarded, except when accompanied by superior empirical demonstration.

B.6 Formalization of the hypothetical method

Before presenting the formula, it is worth putting into words the attitude it condenses. To investigate a hypothesis, in the spirit of this work, is not to try to *prove* that it is true — nor to settle for showing that it “works”. It is to subject it to a twofold movement: on the one hand, to ask how far it makes plausible the evidence we actually observe; on the other, to keep open the possibility that there exists, somewhere in the still-unexplored space of hypotheses, a rival that makes those same pieces of evidence *even more* plausible. The honest scientist must, at one and the same time, defend the hypothesis against the facts and look for whatever might replace it.

Let H be the hypothesis under investigation and E the set of observable evidence. The expression $P(E \mid H)$ reads “probability of observing E assuming that H is true” — that is, it measures how far the hypothesis, if accepted, makes the evidence we in

fact encounter expected. The closer to 1, the more H “explains” E ; the closer to 0, the more the evidence surprises H .²

The hypothetical method, thus understood, is articulated in two simultaneous commitments:

$$\text{Investigation}(H) = \begin{cases} \text{Validate: } H \Rightarrow P(E \mid H) \rightarrow 1 \\ \text{Falsify: } H \Rightarrow \exists H' \text{ such that } P(E \mid H') > P(E \mid H) \end{cases} \quad (\text{B.1})$$

The first line expresses *empirical validation*: showing that, under H , the reproducible evidence E becomes highly expected. It is not a matter of declaring H true in an absolute sense, but of exhibiting that, among the alternatives available today, it is the one that best anticipates what we in fact observe. This is the positive work of the scientist: to deduce consequences from H , compare them with the world, and record the fit.

The second line expresses the *search for a superior hypothesis*: the recognition that there may always exist an H' — not yet formulated, or already formulated and discarded for circumstantial reasons — capable of assigning to the same evidence a plausibility greater than H assigns. This commitment is the antidote to dogmatism. It is not enough that H works well; one must keep asking whether something would work *better*. Every hypothesis, even the most successful, lives under the fertile shadow of its possible supersession.

The two movements are not contradictory, but complementary. To validate without falsifying leads to orthodoxy; to falsify without validating leads to sterile skepticism. To investigate, in the full sense, is to do both at once: to sustain H with the best one can offer and, even so, to keep alive the search for what would render it obsolete.

- **Empirical validation:** to demonstrate that H maximizes the plausibility of E under reproducible evidence.
- **Search for a superior hypothesis:** the commitment to find an H' with greater explanatory power, without dogmas and with openness to revision.

² The word “surprise” is used here instead of “refute” for a technical reason. Refutation, in the strict sense, would correspond to $P(E \mid H) = 0$: the evidence would be *incompatible* with the hypothesis. Scientific practice, however, rarely operates at this categorical limit; what actually occurs is the gradual accumulation of low-likelihood evidence — successive surprises that erode the credibility of H until a rival H' , under which those same pieces of evidence become more probable, prevails. Refutation, in this book, should be understood as the limiting case of surprise, and not as a binary verdict on the truth or falsity of H .

B.7 Boundary examples: hypnotherapy and NDE

B.7.1 Hypnotherapy

Hypnotherapy may be regarded as an intermediate case between protoscience and pseudoscience:

As protoscience: when used on the basis of limited clinical evidence and under professional criteria, representing an initial attempt to build systematic knowledge.

As pseudoscience: when associated with unfounded promises of universal cure or with practices such as “regression to past lives”.

B.7.2 Near-death experiences (NDE)

NDEs illustrate the fine line between science, protoscience, and pseudoscience:

- **Science:** when investigated with methodological rigor (cerebral hypoxia, neurotransmitters).
- **Protoscience:** when still in search of a consolidated theoretical body.
- **Pseudoscience:** when turned into “irrefutable proofs” of life after death.

The boundary between science and pseudoscience depends less on the object itself than on the way it is investigated.

B.8 Charlatanism by imitation and by denial

Pseudoscience and denialism are the fields that most confuse the public, because they assert truths about the world and imitate the aesthetics of science, without any legitimate commitment to evidence or falsifiability.

Table 7 – Comparison between pseudoscience and denialism

Aspect	Pseudoscience	Denialism	Remark
Strategy	Pretends to be science	Attacks consensus	Rhetorical manipulation
Imitates scientific aesthetics	Yes	Yes	Jargon, performative authority
Relation to evidence	Uses false evidence	Rejects real evidence	Selectivity and distortion
Emotional appeal	Mystery, cure	Conspiracy, fear	Cognitive triggers

Unlike denialism and pseudoscience, science fiction, fantasy, and magical realism do not have charlatanism at their core — they are narratives assumed as such.

B.9 Science fiction that got it right

9. Cases in which science fiction anticipated reality

- **Jules Verne** — Journey to the Moon, the electric submarine, communication networks
- **Isaac Asimov** — Autonomous robotics, dilemmas of AI alignment
- **Star Trek** — Cell phone, tablets, voice recognition, telepresence
- **2001: A Space Odyssey** — HAL-9000 anticipated debates about autonomy and trust in AI

When someone reads science fiction, they know it is a matter of narrative supposition, not of factual claim. Its predictive character is not a promise, but, historically, some works anticipated facts with remarkable resonance.

B.10 Hypotheses of *The Metaphor of the Bubbleland*

Many of the hypotheses of *The Metaphor of the Bubbleland* are propositions that have no aim of being proven in the strict sense, but constitute “utopian horizons” — goals that motivate and guide studies, even if the proof is never attained. Others are “working hypotheses”, assumed in order to support scientific hypotheses that will, in turn, be the object of verification.

It should be noted that several of these hypotheses, although individually weak, form a network of interdependent propositions whose internal coherence may confer on the set an epistemic strength greater than that of each isolated part. Section B.3 analyzes in detail the mechanisms and the limits of this mutual reinforcement, and establishes the criteria **Cf1–Cf8** of legitimate strengthening.

The eventual invalidation of a working hypothesis does not necessarily imply the invalidation of the scientific hypothesis it supports. For a complete view of these relations of dependence and support, the reader may consult Appendix AH (Vol. III), which presents a detailed catalog of the hypotheses with their respective classifications and derivations.

B.11 Structurally unfalsifiable hypotheses

There are hypotheses that escape, by construction, the scrutiny of the Scientific Method: they can be neither corroborated nor refuted by any conceivable experiment. The paradigmatic case is the *Hypothesis of the Existence of God*. There is no observational configuration that would make this hypothesis more or less probable in an intersubjectively verifiable way; its acceptance or rejection is a matter of faith, not of evidence.

Hypotheses of this kind violate criterion **Cr5** (refutability in principle) discussed in Section B.2: they are not merely weak hypotheses in the protoscientific sense — they are hypotheses *structurally immune* to any contrary evidence, under any imaginable technology or theoretical framework. They belong, therefore, to the domain of metaphysics or theology, not to that of science.

The distinction is as old as modern natural philosophy itself. It is clearly articulated in the thought of John Locke (Appendix A, Section A.3), for whom faith and rational inquiry operate in functionally distinct domains: the former is a matter of voluntary association and inner experience; the latter demands propositions derived from sensory experience and open to public revision. To confuse the two domains produces dogmatism on one side and scientism on the other — both epistemically harmful.

The appropriate scientific stance toward structurally unfalsifiable hypotheses is not denial (to assert that God does not exist is symmetrical to the opposite assertion and equally unscientific), but *methodological silence*: to recognize that such hypotheses do not belong to the scope of scientific inquiry, without thereby disqualifying them on other planes of human discourse. The scientist, in the capacity of scientist, is skeptical or agnostic by method, not by personal conviction.

This section fulfills a pedagogical function: it establishes, by contrast, what Section B.2 has already articulated in positive terms. A legitimate weak hypothesis is one that, though poorly supported today, remains within the space of potentially testable propositions. A structurally unfalsifiable hypothesis, by contrast, lies outside that space by construction — and no amount of methodological goodwill can change that status.

B.12 Investment in protoscientific programs

Oftentimes, the distance between hypothesizing and realizing is measured in thousands or millions of dollars.

— ANTONIO FERRÃO NETO

No hypothesis enters the scientific corpus without passing through the scrutiny of the Scientific Method. Before any validation, however, it is necessary to invest time, human resources, and funding in its exploration. A practical question then arises that every protoscientific work is obliged to confront: *under what circumstances is it rational to allocate resources to a research program whose hypotheses have not yet been empirically validated?*

The question is not speculative: it has an answer in the consolidated philosophy of science of the twentieth century, articulated in two complementary layers.

B.12.1 The Lakatosian criterion: progressive versus degenerative programs

Imre Lakatos (Appendix A, Subsection A.4.4) distinguished two possible trajectories for a research program. A program is **progressive** when it generates new predictions, opens up unexpected connections between previously disconnected domains, and anticipates phenomena that come to be observed. It is **degenerative** when it limits itself to accommodating, by means of *ad hoc* auxiliary hypotheses, already-known facts without anticipating any new fact.

For Lakatos, the rationality of investing in a research program does not depend on the definitive truth of its hard core — inaccessible at any stage — but on its *trajectory*: whether it is generating new and non-trivial consequences or merely defending itself against accumulated anomalies.

Applied to the *Bubbleland Metaphor*, the criterion operates as follows. The 34 weak hypotheses cataloged in Table 6 do not constitute isolated propositions, but an interdependent network that produces new predictions in distinct domains:

- **Molecular biology:** tunneling of biomolecules into a fourth spatial dimension (H14, H15, H38) predicts specific experimental signatures in conformational transitions not yet mapped.
- **Particle physics:** the hypothesis of solitons as the ontological basis of the Standard Model (H40–H44) predicts that the 19 free parameters of the SM must be determined by the internal geometry of the solitons.

- **Cosmology:** dark matter and dark energy as inter-reality nomological phenomena (H36, H37) predict patterns of correlation between cosmological observables today treated as independent.
- **Quantum mechanics:** temporal detuning between parallel realities (H19, H20) predicts measurable deviations in radioactive decays under conditions of extreme shielding.

Each of these predictions is technologically distant; none is structurally inaccessible. The program, in the Lakatosian sense, satisfies the criterion of progressiveness — not because it has already generated confirmations, but because it generates *differential predictions* in multiple domains.

B.12.2 Formalization of the expected value under epistemic uncertainty

The Lakatosian criterion gives the qualitative form of the decision; it remains to give it a quantitative form. Decision theory under epistemic uncertainty offers the following minimalist formalization. Let H_i be one of the weak hypotheses of the program, p_i its subjective probability of eventual confirmation, V_i the expected scientific value of its confirmation, and C the total cost of the research program. The expected value of the investment is:

$$\mathbb{E}[\text{Investment}] = \sum_{i=1}^N p_i \cdot V_i - C \quad (\text{B.2})$$

The rationality of investing corresponds to the condition $\mathbb{E}[\text{Investment}] > 0$. Three observations about this formalization are essential.

First: the probabilities p_i are not known.

In the protoscientific regime, p_i is not robustly estimable — and this is precisely the characteristic that distinguishes protoscience from mature science. This does *not* invalidate the formula, but redirects its use: it serves as a conceptual organizer, not as a calculator. The operational question becomes “what is the plausible lower bound of p_i below which the investment ceases to be rational?”, and not “what is the exact value of p_i ?”.

Second: the value V_i is strongly asymmetric across hypotheses.

The confirmation of some hypotheses of the program — for example, 4D biomolecular tunneling (H14) or the nomological structure of dark matter (H36) — would have transformative scientific value, opening up entirely new domains of research. The confirmation of others would have more modest value. It is the asymmetry of V_i that justifies investing even when p_i is low: not because the gain is infinite (it is not), but because the potential gain is finite and sufficiently large.

Third: the cost C is distributed and partially reversible.

Protoscientific programs do not require massive initial investment; they can be explored at an adjustable scale, with C allocated in increments whose partial return can be evaluated before larger investments. This staged structure distinguishes rational investment in protoscience from massive investment in blind bets.

B.12.3 Synthesis

The combination of the two criteria — qualitative Lakatosian progressiveness and quantitative expected value under uncertainty — justifies investment in programs such as that of this work without invoking arguments of infinite payoff or theological appeals. The rationality of the bet rests on three articulated points:

1. The program is progressive in the Lakatosian sense: it generates new predictions in multiple domains.
2. The expected scientific value of partial confirmations is high, albeit finite.
3. The cost is scalable, allowing incremental evaluation before larger commitments.

These criteria are public, auditable, and independent of the personal conviction of any agent. They therefore differ from the decision structure of *Pascal's Wager*, in which the investment is justified by infinite payoff under a binary and untestable hypothesis — a structure that, although historically important, does not apply to programs of empirical research.

In particular, the structure defended here is *indistinguishable in form* from the criterion applied to any frontier program in theoretical physics (string theory, loop quantum gravity, supersymmetry), in which vast investments have been allocated for decades on the basis of inestimable p_i and potentially transformative V_i . The

difference between those programs and the *Bubbleland Metaphor* is not structural — it is merely one of stage of institutionalization.